

Sending mixed signals: convergent iridescence and divergent chemical signals in sympatric sister-species of Amazonian butterflies

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eLife Assessment

This study presents a **valuable** assessment of increased similarity in visual appearance combined with an increased chemical difference between two butterfly species in sympatry compared with differences between three populations of one of the two species in allopatry. While the evidence is **solid**, its interpretation in terms of evolutionary responses to shared predators (visual signals) and avoiding between-species mating (chemical signals) is overstated due to the lack of direct experimental evidence.

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Abstract

Closely-related species are often partitioned into different geographic areas or micro-habitats as a result of competition and reproductive interference. Here, we investigate how the evolution of shared adaptive traits may shape ecological interactions and genetic divergence, and facilitate co-existence in sympatry. In closely-related *Morpho* butterfly species living in the understory of the neo-tropical rainforest, the blue iridescent coloration of the wings is likely involved in predation evasion as well as in mating recognition and courtship. These contrasted selective pressures acting on this visual trait shared between closely-related species has likely shaped their coexistence. We used spectrophotometry, behavioral experiments, visual modeling and chemical analyses applied to samples from allopatric and sympatric populations of *Morpho helenor* and *Morpho achilles* to characterize how the evolution of visual and chemical traits might favor their coexistence in sympatry: we quantified the differences in wing iridescence and tested for variations in the sexual preference for this trait between allopatric vs. sympatric populations. We found a strong

similarity in iridescence among species in sympatry, suggesting convergence driven by predation. Although behavioral results suggest that iridescent signals could also be used as visual cues during mate choice, convergent iridescent signals may impair the visual recognition of sympatric *Morpho* species. In contrast, divergent chemical bouquets among species suggest that the visual similarity of sympatric *Morpho* species might have favored the divergence of alternative traits involved in species recognition such as chemical cues. This study underlines how ecological interactions and trait evolution can shape species coexistence in sympatry.

Introduction

Identifying the evolutionary factors promoting diversification and co-existence of closely-related species in sympatry is a major challenge. Since the evolution of traits within species in sympatry can affect ecological interactions between individuals, it either fuels or impairs the speciation process. Closely-related species often share similar suites of traits because of their shared evolutionary history. Furthermore, in sympatry, habitat filtering excludes species with suites of traits mal-adapted to local conditions, therefore enhancing trait similarity (Chazot et al., 2014 [DOI](#)). The shared selective pressures encountered by individuals living in sympatry also often promote the evolution of locally adapted traits within species and even trait convergence among species (Muschick et al., 2012 [DOI](#)). That said, closely-related sympatric species often display divergent suites of traits because they are usually partitioned in divergent ecological niches (Berlocher & Feder, 2002 [DOI](#)). The co-existence in sympatry may also promote divergent evolution of traits in incipient species, because of competitive interactions between individuals or as a result of reinforcement (Butlin & Smadja, 2018 [DOI](#)). The evolution of traits in incipient species living in sympatry thus depends on the balance between convergence and divergence, resulting either in (i) evolutionary trade-offs of traits jointly influenced by antagonistic natural and sexual selective pressures (Ellers & Boggs, 2003 [DOI](#); Maan & Seehausen, 2011 [DOI](#)) or (ii) convergence in some traits and divergence in others (Le Roy, Roux, et al., 2021 [DOI](#)).

Theoretical and empirical studies have shown that sexual selection may favor the evolution of preferences for locally-adapted traits within species (Servedio, 2004 [DOI](#); Servedio & Boughman, 2017 [DOI](#); van Doorn et al., 2009 [DOI](#)). For instance, the predator-deterrent coloration of poison frogs is also detected and used as mating cues by females (Reynolds & Fitzpatrick, 2007 [DOI](#)). Similarly, habitat-dependent coloration of sympatric cichlid fish is also used as a visual cue for mate recognition (Seehausen et al., 2008 [DOI](#)). Yet, sexual interactions are likely to occur between individuals from closely-related species when they live in sympatry, and similar preferences for adaptive traits may thus result in substantial reproductive interference (Gröning & Hochkirch, 2008 [DOI](#)). For example, because sympatric amphibians can sometimes share the same mating cues and preferences, they often engage in misdirected mating attempts towards heterospecifics, a potentially deleterious behavior (Soni et al., 2024 [DOI](#)). In addition, hybrids, when produced, can be unfit, thus favoring the evolution of sexual preferences for species-specific cues, rather than locally-adapted traits (Maisonneuve et al., 2024 [DOI](#)). Comparing variations in adaptive traits in sympatric vs. allopatric populations of recently-diverged species, and testing the sexual preference for those traits can shed light on the selective processes targeting traits modulating reproductive isolation and co-existence in sympatry.

In butterflies, the evolution of wing color patterns can be influenced by both natural and sexual selection. The visual discrimination of wing color pattern has been shown to enable intraspecific recognition during courtship in many species (Costanzo & Monteiro, 2007 [DOI](#); Li et al., 2017 [DOI](#)). However, the evolution of wing color patterns is also strongly influenced by the visual effect perceived by predators (Finkbeiner et al., 2014 [DOI](#); Oliver et al., 2009 [DOI](#)). Whether these selective pressures promote trait convergence or divergence in sympatric species might depend on their relatedness: for instance, a recent study in *Papilionidae* showed multiple color pattern

convergences between distantly-related species living in sympatry, while divergent colorations are found in closely-related species (Puissant et al., 2023 [DOI](#)). Positive selection acting on divergent traits involved in species recognition could be increased because of higher reproductive interference in closely related species than in distantly related taxa (Pfennig & Pfennig, 2009 [DOI](#)). In sympatric species with chemical-defenses, such as *Heliconinii* butterflies, local predation pressures tend to promote the convergence of similar conspicuous warning wing patterns (i.e. Müllerian mimicry, (Joron et al., 1999 [DOI](#); Merrill et al., 2014 [DOI](#))). But the costs associated with hybrid production in turn favor the evolution of alternative divergent mating cues in mimetic butterflies (Estrada & Jiggins, 2008 [DOI](#)), and divergence in male pheromone bouquets and female attraction has been found among mimetic sister species (González-Rojas et al., 2020 [DOI](#)). Similarly, the evolution of specific visual mate recognition signals, limiting reproductive interference but indistinguishable by predators, can also be promoted on the wings of mimetic butterflies (Laurens et al., 2014 [DOI](#)). Thus, ecological interactions among recently-diverged species living in sympatry can impact the evolution of traits involved in both local adaptation and mate recognition and may trigger the evolution of alternative traits. Investigating variations in traits used as mating cues in allopatric vs. sympatric populations of sister-species can help ascertain how trait evolution might favor speciation and co-existence in sympatry.

Here we focus on the evolution of mating cues in the neotropical butterfly genus *Morpho*, where multiple closely-related species co-exist in sympatry (Blandin & Purser, 2013 [DOI](#)). In some *Morpho* species, striking iridescent blue coloration is displayed on the dorsal side of the wings, due to nanostructures on wing scales (Siddique et al., 2013 [DOI](#)). Variations in the reflected wavelength depending on the angle of observation and/or illumination (i.e. iridescence) have been documented in *Morpho* species (Giraldo et al., 2016 [DOI](#)). The light signal reflected by iridescent surfaces can be very directional, as hue and brightness of iridescent objects can drastically change depending on the light environment or the observer's position (Doucet & Meadows, 2009 [DOI](#)). Directional iridescent signals generated in animals could then enhance recognition by mates while remaining poorly detected by predators (Endler, 1992 [DOI](#)). For instance, in some hummingbird species, when males tilt their head in a specific direction, flashes of bright coloration can be observed in these otherwise darkly-colored birds (Simpson & McGraw, 2019 [DOI](#)). The specific directional signal produced by the iridescent feathers is a cue during courtship, suggesting that the antagonistic sexual and natural selective pressures finely shape the evolution of iridescence. Similarly, male *Hypolymnas* butterflies display abrupt shifts in UV brightness on their wings and their courtship behavior enhances the production of flashes perceived by females (White et al., 2015 [DOI](#)). How much sexual selection shapes the evolution of iridescence in sympatric *Morpho* species is currently unknown, but behavioral experiments carried out in the field in Amazonian Peru highlighted strong visually- based territorial interactions among males from sympatric species and limited species discrimination based on female coloration in males (Le Roy, Roux, et al., 2021 [DOI](#)).

This raises questions on the key visual cues involved in mate choice, given that the iridescent blue coloration shared by closely-related species encountered within the understory is likely selectively constrained by predators. The iridescent bright blue dorsal coloration of *Morpho* wings contrasts with the brown and matte ventral side, and generates a peculiar visual effect during flight. The combination of the blue flashes produced by the alternate exposure of the bright blue vs. brown sides of the wings during flapping flight, in addition to erratic flight trajectories makes these *Morpho* very difficult to follow by predators (Young, 1971 [DOI](#)), potentially enhancing their evasive capabilities (Murali & Kodandaramaiah, 2020 [DOI](#); Vieira-Silva et al., 2024 [DOI](#)). Experimental trials with evasive prey have shown that predators learn to avoid prey they repeatedly fail to catch (Páez et al., 2021 [DOI](#)). The display of iridescent wings could thus be associated with a higher survival rate in nature because of both (i) direct effects, through successful escape of predator attacks, and (ii) indirect effects, by limiting predation attempts by birds recognizing the blue signals and refraining from attacking (Pinheiro et al., 2016 [DOI](#)). This indirect effect could promote the evolution of convergent blue patterns in sympatric species, similar to the mimicry observed in

species with chemical-defenses. In line with this hypothesis, repeated local convergence in the proportion of iridescent blue vs. black areas on the dorsal side of the wings have been documented in the sister-species *Morpho helenor* and *Morpho achilles* living in sympatry throughout the Amazonian basin: indeed, parallel geographic variations in the width of the dorsal blue band was detected in these sympatric species (Laurens et al., 2021 [DOI](#)). Precise quantification of variations in iridescence is now needed to assess the respective effects of selection by predators and mates, that may drive convergent vs. divergent evolution of iridescence in sympatric species sharing the same micro-habitats. Investigating this convergent vs. divergent evolution of visual cues that may act as pre-zygotic barriers to gene flow between these two species will shed light on the mechanisms allowing their genetic divergence and coexistence in sympatry.

To do so, we quantified variations of iridescence in the two sister species *M. helenor* and *M. achilles* in different geographic localities to investigate the antagonistic effects of selective pressures generated by mate recognition and shared predation. We used behavioral experiments to test for the effect of this color variation in visual signals in mate preferences. Finally, we studied variations in the volatile compounds produced by wild males and females from both species to explore the evolution of potentially alternative traits, such as chemical cues, possibly acting as a reproductive barrier and favoring the coexistence in sympatry of closely related *Morpho* species.

Results

We investigated differences in wing iridescence, color perception and color pattern-based mate choice of two sister species of *Morpho*, *M. helenor* and *M. achilles*, which diverged ~ 3.6 Mya (Chazot et al., 2021 [DOI](#)). We relied on geographic variations of wing color patterns within *M. helenor* (**Fig 1** [DOI](#)) to compare the iridescence of three different populations of *M. helenor* (subspecies *M. helenor bristowi*, *M. helenor theodorus* and *M. helenor helenor*) and test the associated visual effect using visual models.

The convergent iridescent signals of sympatric *Morpho* species are not differentiable

We measured the reflectance of *Morpho* wings at different angles of observation and illumination, in two different planes (referred to as proximo-distal and antero-posterior planes) following two complementary protocols (referred to as “specular” and “tilt”; see **Fig 2** [DOI](#).A and Supporting Information Text 1 for extended methods). We then extracted for each spectrum three commonly-used colorimetric variables: Hue, Brightness and Chroma (analyzed in S4 Fig).

We found that hue and brightness always vary depending on the angle of incidence of the light, as confirmed by PERMANOVAs detecting a significant effect of the “Angle” for all methods of measurements (**Fig 2** [DOI](#) and statistical analyzes in S1 Table and S5 Fig) confirming that in both species, both sexes are iridescent. Second, the brightness of *Morpho* wings is highly sexually dimorphic, as the effect of sex is always significant. The brightness measured with the “tilt” protocol (**Fig 2** [DOI](#).F and 2.G) shows that males tend to be brighter than females in both Ecuador and French Guiana.

To finely appraise the complete spectrum of variation in iridescence within and among *Morpho* species, we then used a multivariate approach allowing to analyze the full reflectance spectra obtained from both proximo-distal and anteroposterior measurements. At the intraspecific level, significant divergence of iridescence between populations was detected (**Fig 3** [DOI](#).B). In contrast, iridescence did not differ between species living in sympatry (PERMANOVA, *p*-value = 0.084),

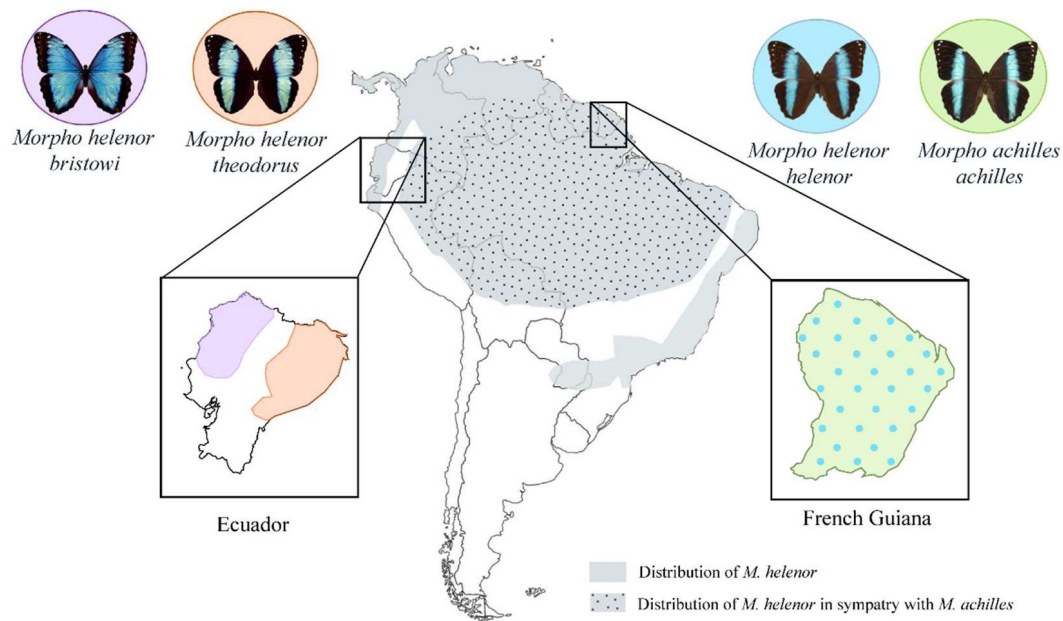


Fig 1

Distribution of the sister-species *M. helenor* and *M. achilles* across South America.

The gray area represents the whole distribution of *M. helenor*, and the dotted area indicates the localities where *M. helenor* is in sympatry with *M. achilles*. Note that the different subspecies of *M. helenor* are found in different localities and display substantial variations in the proportion of black vs. blue areas on the dorsal sides of their wings. For example, *M. h. bristowi* and *M. h. theodorus* are allopatric in Ecuador, with their respective distribution separated by the Andes such that they are never in contact. While *M. h. bristowi* is found on the Pacific side of the country and has a wide blue band on the dorsal side of its wings, *M. h. theodorus* is found in Western Amazonia and displays a narrow blue band on the dorsal side of its wings. However, *M. helenor* is sympatric with *M. achilles* throughout the Amazonian rainforest. In French Guiana, the subspecies *M. helenor helenor* and *M. achilles achilles* display convergent dorsal color patterns with thin blue bands (Llaurens et al., 2021 [\[1\]](#)).

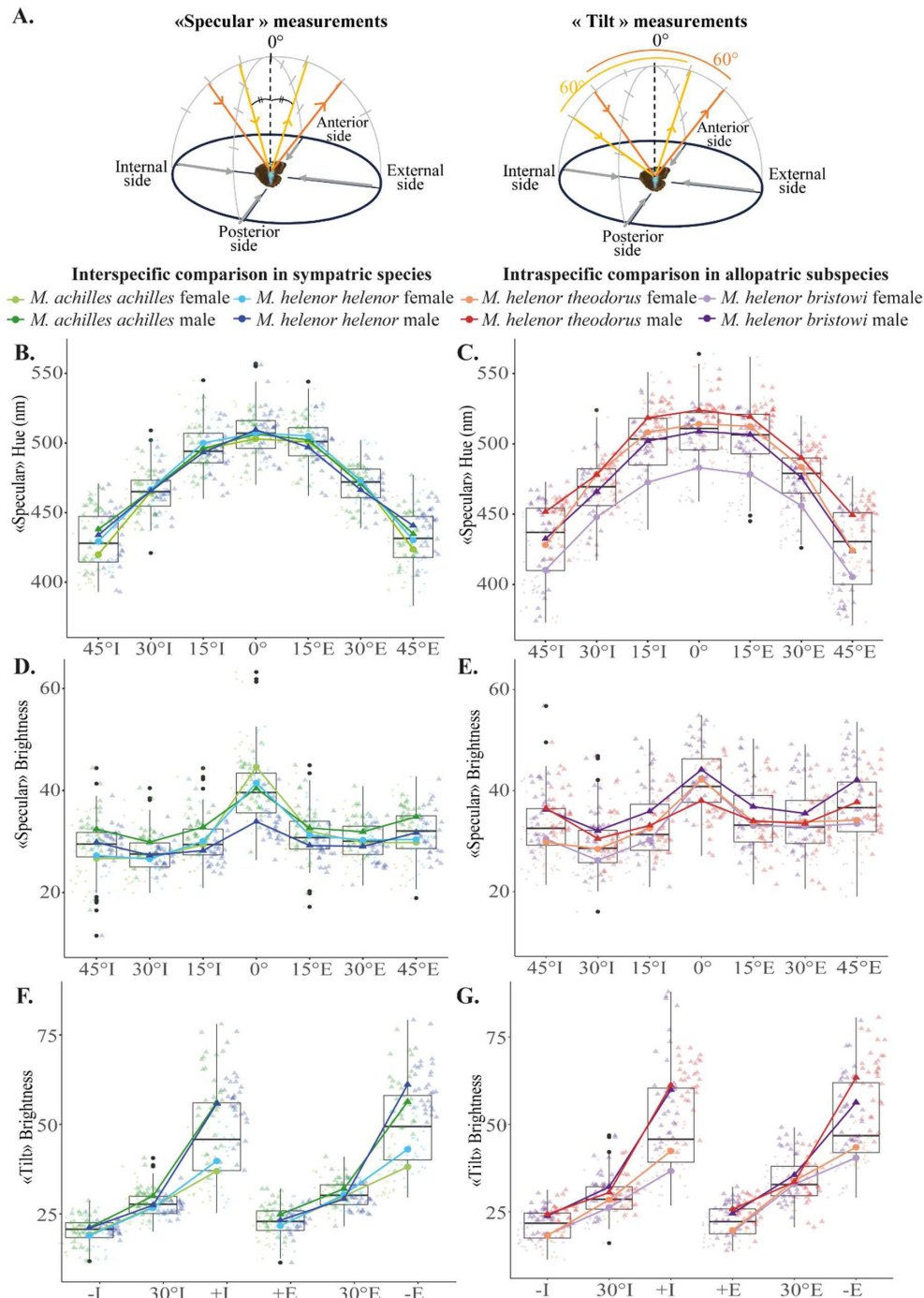


Fig 2

Differences of hue and brightness on the proximo-distal plane of *Morpho* wings

(A) Illustration of the two protocols used to assess the differences in *Morpho* wing reflectance. The “Specular” set-up allows for the quantification of the wing color variations, while the “tilt” set-up can be used to quantify brightness variation for each *Morpho* wing (see Appendix for extended methods). Variations of hue (B and C) and brightness (D and E) calculated from the wing reflectance measured with the “Specular” set-up, and variations of brightness calculated from the wing reflectance measured from the “tilt” set-up (F and G). Those hue and brightness parameters were calculated for the sympatric *M. h. helenor* and *M. a. achilles* (first column in green and blue) and for the allopatric *M. h. theodorus* and *M. h. bristowi* (second column in orange and purple) on the proximal-plane plane of their wings (I= illumination on the internal side of the wings, E= illumination on the external side of the wings). See S1 Table for the PERMANOVA analyses describing those graphs.

suggesting that iridescence converges in sympatry (**Fig 3** [A](#)). In both cases, the interaction term *taxon*sex* was non-significant, suggesting that the sexual dimorphism of iridescence was similar for all taxa.

We then modeled the perception of color contrast in the visual systems of *M. helenor* and UV-sensitive birds respectively, and tested whether the differences of reflectance observed between sexes and species at different angles of illumination could potentially be perceived by a *Morpho* observer or a putative predator. As sexual dimorphism on wing reflectance was detected, we measured the chromatic distances for males and females separately (**Fig 3** [C](#)). As the results were similar along the two directions of measurements (antero-posterior and proximo-distal), only proximo-distal results are shown below (antero-posterior results in S6 Fig). *Morpho* butterflies and predators can visually perceive the difference in the blue coloration between the allopatric populations of *M. helenor*: for both *Morpho* and bird visual systems the mean chromatic contrasts are always higher than 1 JND, for both sexes and for all angles. In contrast, the mean chromatic contrasts measured between sympatric species were below 1 JND: the minute differences observed between sympatric species are thus unlikely to be discriminated either by *Morpho* or bird predators.

Evidence of mate preferences based on visual cues within species

We then performed a series of mate choice experiments to identify the visual cues used during mate choice within *M. helenor*. We specifically tested for preferences based on the differences in color measured between the two previously studied populations of *M. helenor* (*M. h. bristowi* and *M. h. theodorus*), which significantly differs in both wing pattern and iridescence.

We first conducted a tetrad experiment using the allopatric subspecies *M. h. bristowi* and *M. h. theodorus*. The highest number of mating events occurred between the two assortative pairs (see **Table 1** [A](#)). More than half of the matings occurred between a female and a male of the *M. h. bristowi* population. Indeed, a Chi-square test performed on the contingency table showed a significant departure from random mating ($X^2 = 4.98$, $p\text{-value} = 0.04$), confirmed by a Fisher exact test ($p\text{-value} = 0.04$). The time until an assortative mating ($\text{mean} = 48$ min, $\text{sd} = 38$ min) or until a disassortative mating ($\text{mean} = 48$ min, $\text{sd} = 38$ min) however did not significantly differ ($T\text{-test}$; $\text{stat} = -0.27$, $p\text{-value} = 0.79$).

Different pairs of female dummies were presented to the *M. helenor* males to investigate assortative preference between (sub)species (experiment 1), discrimination of the wing pattern (experiment 2) and discrimination of the iridescent coloration (experiment 3) (see the experimental details in the M&M section).

Experiment 1 tested the male assortative visual preference using two dummies each harboring the wings of a female of a different *M. helenor* subspecies. *M. h. theodorus* males did not show any preference for any dummy female, as shown by the probability of approaches and touches that were not different from the random 0.5 expectation (approaches: Wilcoxon test, $p\text{-value} = 0.764$, **Fig 4** [A](#); touches: Wilcoxon test, $p\text{-value} = 0.287$, **Fig 4** [B](#)). *Morpho helenor bristowi* males, in contrast, approached and touched a con-subspecific *M. h. bristowi* dummy female significantly more often than expected by chance alone (approach: Wilcoxon test, $p\text{-value} = 0.0448$, **Fig 4** [A](#); touch: Wilcoxon test, $p\text{-value} < 0.001$, **Fig 4** [B](#)), suggesting a preference toward the *M. h. bristowi* females based on the recognition of visual cues alone.

Experiment 2 tested male pattern-based discrimination, using two dummies harboring the wings of *M. h. bristowi* females (i.e. same iridescence), but one of which was modified so as to present the thin blue bands found in *M. h. theodorus* (i.e. a different pattern). *M. h. bristowi* males tended to approach and touch significantly more often the wild-type *M. h. bristowi* female model as compared to the modified *M. h. bristowi* female model exhibiting a *theodorus* pattern (Wilcoxon tests on approaches; $p\text{-value} = 0.0188$, on touches; $p\text{-value} < 0.001$, see S7 Fig). As both female

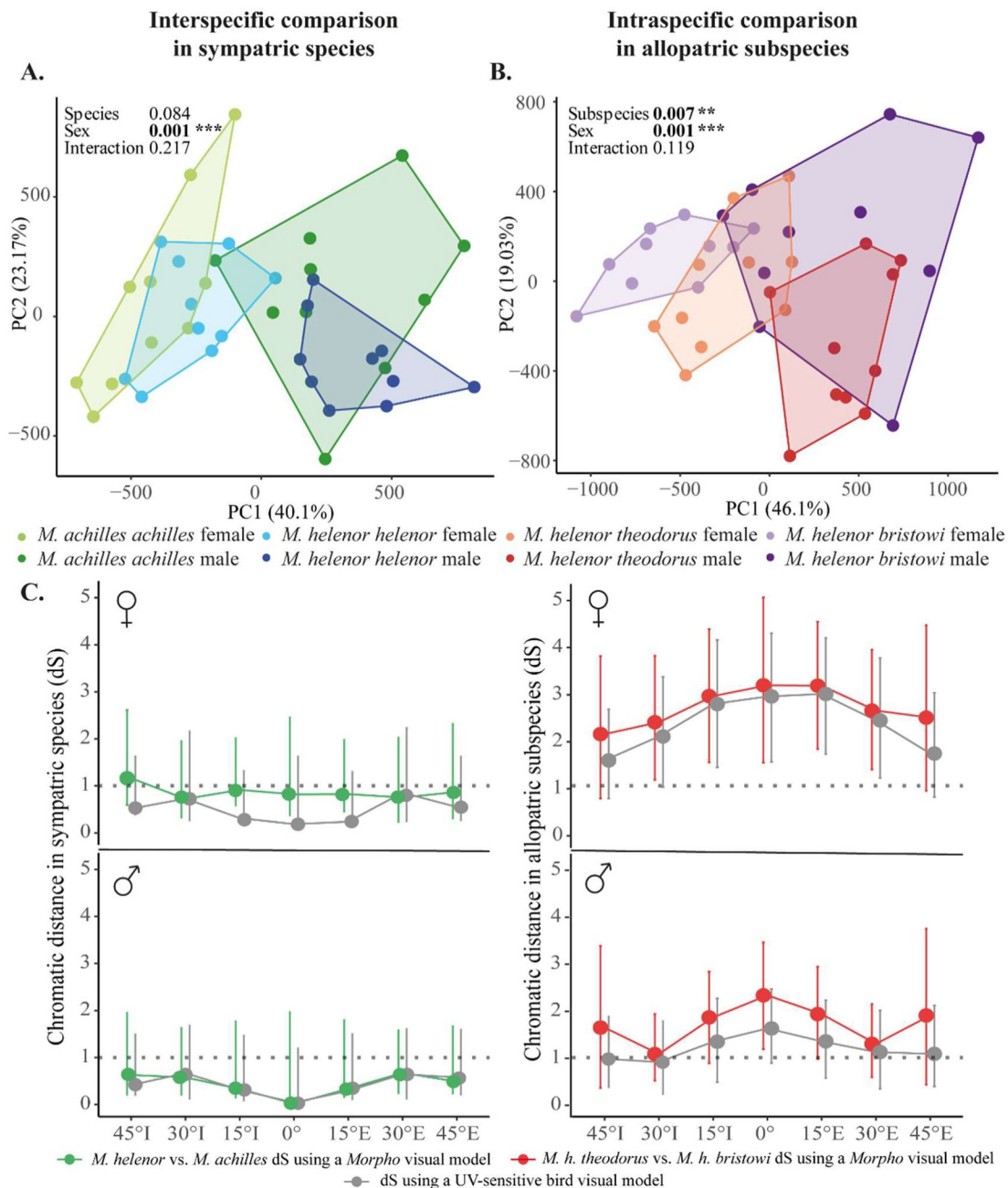


Fig 3

Characterization and perception of the iridescent coloration of *Morpho* butterflies.

(A) and (B): PCAs showing the variation in iridescence for both sexes, (A) in the two sympatric species from French Guiana (*M. h. helenor* vs. *M. a. achilles*) and (B) in the two allopatric Ecuadorian subspecies of *M. helenor* (*M. h. theodorus* vs. *M. h. bristowi*). Each point represents the global signal of iridescence of each individual, corresponding to the 21 complete reflectance spectra obtained from the 21 tested angles of illumination. The results of the PERMANOVA are shown on the top left corner of each graph. (C) The chromatic distances (*i.e.* the visual discrimination rate by a visual model) of the wing reflectance measured with the “Specular” set-up on the proximo-distal plane. Visual modeling was used to calculate the chromatic contrast of blue coloration between allopatric *M. helenor* subspecies (red) and between the two sympatric sister-species *M. helenor* and *M. achilles* (green), as perceived by a *Morpho* visual system for every angle of illumination measured on the proximo-distal plane. The chromatic contrast likely perceived by UV-sensitive birds is shown in gray. Chromatic contrast of the female wings (top) and male wings (bottom). The threshold of discrimination is shown by the dotted line and set to 1 *Just Noticeable Difference* (JND).

N=30

	M. h. theodorus female	M. h. bristowi female
M. h. theodorus male	6	3
M. h. bristowi male	5	16

Table 1

Tetrad experiment results:

Number of mating events involving the different possible pairs of males and females from the two subspecies of *M. helenor* (bristowi vs. theodorus) in 30 tetrad experiments

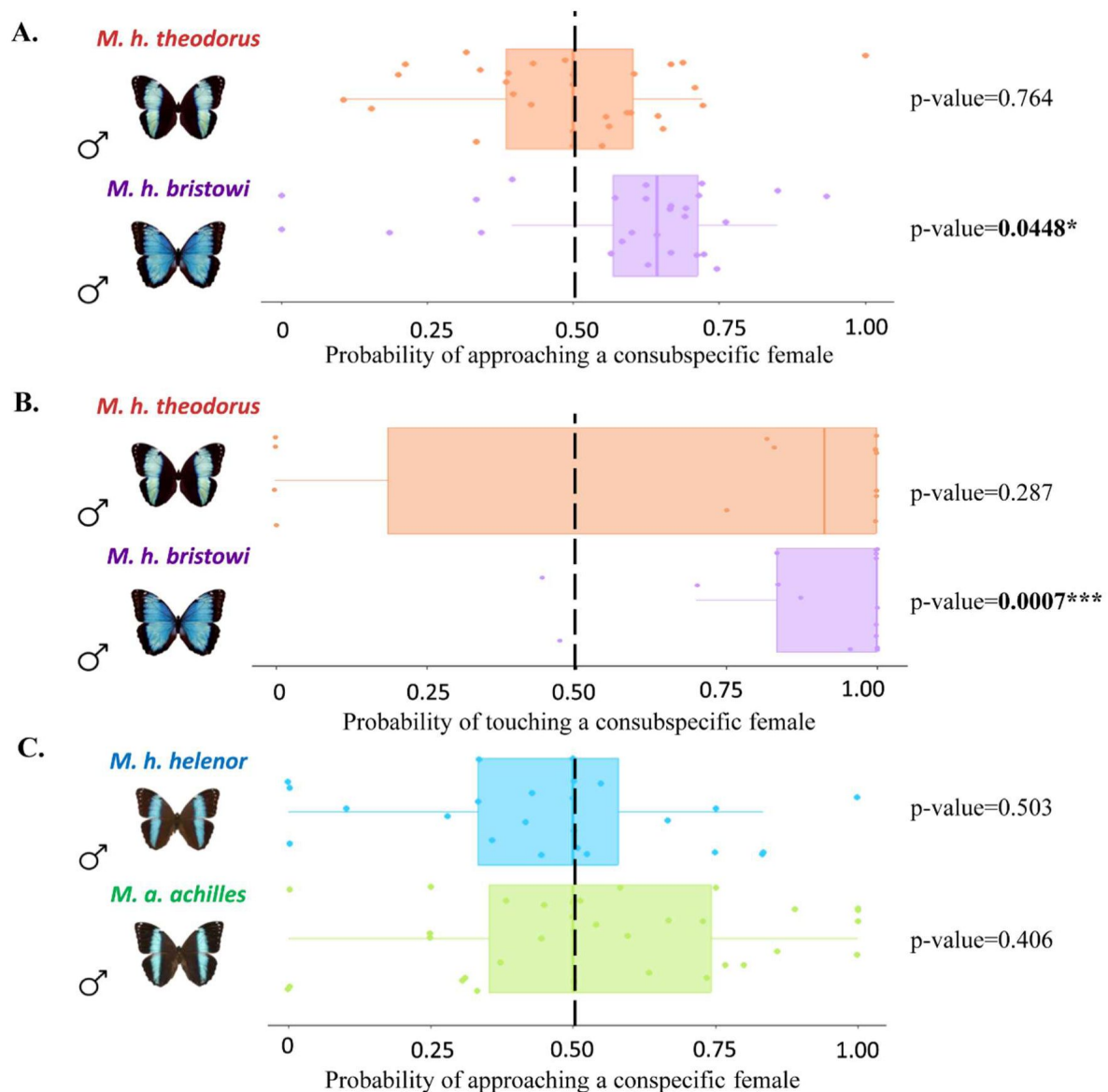


Fig 4

Morpho male preference based on visual cues alone.

Probabilities of (A) approaching and (B) touching a con-subspecific female for *M. h. bristowi* males (purple) and *M. h. theodorus* males (orange) from Ecuador, as well as the probabilities of (C) approaching a con-specific female for *M. h. helenor* males (blue) and *M. a. achilles* males (green) from French Guiana, measured during *Experiment 1*. The dotted line indicates the expected probability of approaching/touching a model if no preference is present. The *p-values* of Wilcoxon tests testing for the significant departure from the 0.5 probability are shown next to the corresponding graphs.

dummies shared the same blue iridescence, the choice of *M. h. bristowi* males was likely influenced by the difference in the pattern displayed by the dummy females, suggesting that the color pattern is used by males during mate choice.

Experiment 3 tested male color-based mate discrimination using two dummies harboring the wings of females from the two subspecies (i.e. different iridescence), but those of *M. h. bristowi* were modified so as to resemble *M. h. theodorus* (i.e. same pattern). *Morpho helenor theodorus* males tended to approach and touch significantly more often the *M. h. theodorus* female model compared to the artificially modified, theodorus-like *M. h. bristowi* female dummy (Wilcoxon test on approaches; $p\text{-value} = 0.0201$, on touches; $p\text{-value} = 0.0343$, see S8 Fig). As both dummy females shared the same color pattern, the preference expressed by *M. h. theodorus* males likely stems from the differences in the iridescent blue color suggesting color-based discrimination in males. Nevertheless, in experiment 2 and 3, modified dummies were strongly rejected by males. We cannot entirely rule out that the artificial modifications performed on our dummies could generate a specific visual rejection by the males.

Visual cues are not pre-zygotic reproductive barriers between the mimetic species M. h. helenor and M. a. achilles In a final experiment (experiment 4), we tested whether males from the two sympatric species from French Guiana (*M. achilles* and *M. helenor*) have a visual preference toward their conspecific females: two female dummies harboring the wings of females from the two species were presented to each male. While none of the males (from either *M. helenor* or *M. achilles* species) touched the female models, they did approach them in the cages. The approach rate and the number of approaches did not significantly differ between *M. helenor* males (rate = 61%; mean = 4.21, sd = 5.34) and *M. achilles* males (rate = 66%; mean = 7.18, sd = 8.29; binomial GLM: $p\text{-value} = 0.626$), revealing similar levels of male courtship motivation. The independent analysis of the mean preference per species shows that neither *M. helenor* (Wilcoxon test, $p\text{-value} = 0.503$) nor *M. achilles* (Wilcoxon test, $p\text{-value} = 0.406$) males approached significantly more conspecific females as compared to the hetero-specific females in the cage (Fig 4C), suggesting that the difference among the convergent color patterns and blue iridescence of female *M. helenor* and *M. achilles* is not perceived by males.

Divergent chemical profiles observed in males between the mimetic species *M. helenor* and *M. achilles*

We finally investigated the volatile chemical compounds extracted from the genitalia of males and females from the mimetic sympatric species *M. helenor* and *M. achilles*. The nMDS analyses performed (Fig 5) suggest a strong divergence in the chemical profiles between males of the two species, especially for C16 to C30 compounds. This result contrasts with the strong similarity of chemical profiles found between the females of these two species in these longer-chain compounds (S9 Fig).

The PERMANOVA of the chemical spectra of *M. h. helenor* and *M. a. achilles* genitalia revealed a significant effect of sex, species ($p\text{-value} < 0.001$ on both the C8 to C16 and the C16 to C30 datasets, see Fig 5) and a significant interaction between the two variables. When analyzing the results for each sex separately (post-hoc), we found a significant effect of the species variable (PERMANOVA, $p\text{-value}=0.001$ for males and $p\text{-value} = 0.049$ for females in the C8 to C16 compounds, PERMANOVA, $p\text{-value}=0.001$ for males and $p\text{-value} = 0.048$ for females in the C16 to C30 compounds), suggesting a substantial divergence between species in chemical bouquets, especially marked in males.

Next, we performed an Indicator Value Analysis to identify the different compounds significantly associated with each species, separating males and females (Table S2). Among all the annotated compounds, we found traces of beta-ocimene, previously identified as a butterfly pheromone.

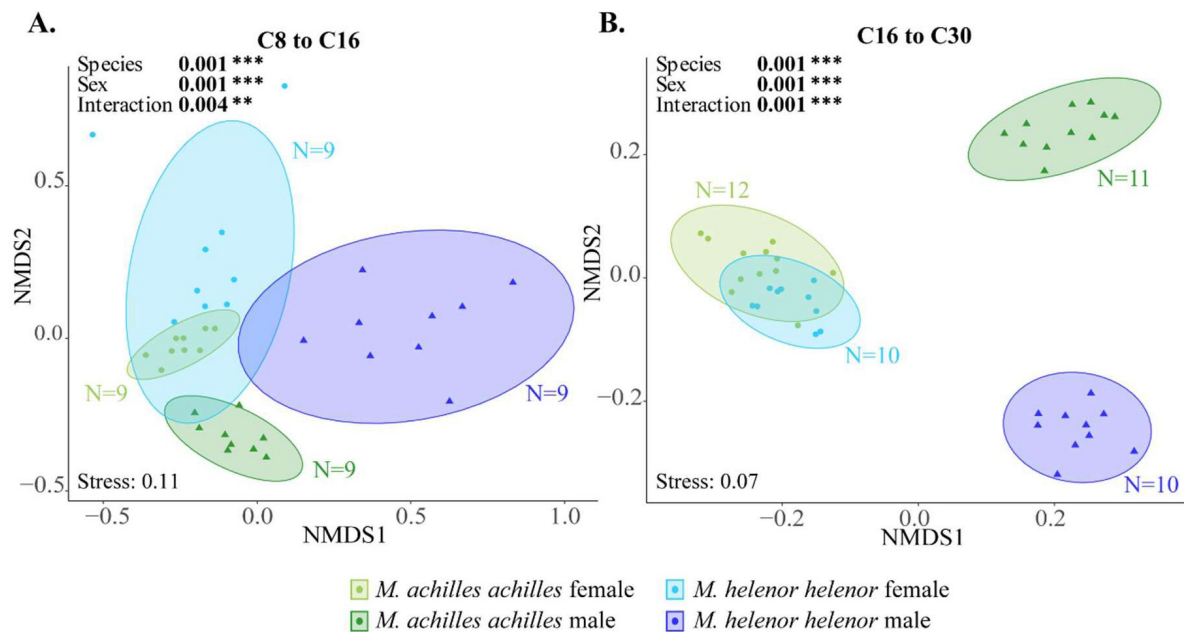


Fig 5

Divergent chemical compounds found on the genitalia of sympatric *Morpho* males.

nMDS representation of the differences in the (A) C8 to C16 chemical compounds and (B) C16 to C30 chemical compounds found in the genitalia of *M. h. helenor* and *M. a. achilles* males and females, calculated using Bray-Curtis distances. *M. h. helenor* are shown in blue (males are in dark blue and females in light blue), and *M. a. achilles* are shown in green (males are dark green and females are light green). The result of the PERMANOVA (999 permutations) testing the effect of sex and species on the chemical composition of *M. h. helenor* and *M. a. achilles* are shown on each figure.

Interestingly, different proportions of this compound were found on the genitalia of *M. helenor* and *M. achilles* males.

Discussion

Convergent iridescence between sympatric *Morpho* butterflies suggests a prominent role of predation on the evolution of iridescence

Our study revealed a striking similarity in the iridescent coloration of the blue patches in the sympatric species of *M. helenor* and *M. achilles* in French Guiana, in sharp contrast with the divergent iridescence observed between allopatric populations of *M. helenor* sampled in different geographic areas. Together with the convergent evolution of the blue band width previously detected in these sympatric species (Llaurens et al., 2021 [DOI](#)), this observation is consistent with there being a primary role of visual predation on the evolution of wing coloration in *Morpho* (Chouteau et al., 2016 [DOI](#); Stuckert et al., 2014 [DOI](#)). The results of the visual models showing that these convergent iridescent signals are likely undistinguishable by avian predators further supports there being positive selection generated by predators promoting mimicry in these sympatric species. These results suggest that predation pressure can strongly impact the evolution of iridescence, and are further supported by the lack of directionality for the iridescent signal generated by *Morpho* wings. Directionality in iridescent signals is expected when sexually selected traits incur a predation cost due to their conspicuousness: directionality might enhance signaling to mates while remaining poorly detected by predators (Endler, 1992 [DOI](#)). Our study shows that *M. helenor* and *M. achilles* wings display a gradual variation of hue and brightness when changing the illumination/observation angles contrary to some other iridescent butterfly species (e.g. *Hypolimnas bolina*; (White et al., 2015 [DOI](#))). The lack of signal directionality observed in these *Morpho* species thus suggests that the iridescent signal is likely on display for both potential mates and predators to see. These gradual changes in hue and brightness could also enhance the confusing effect generated by the alternate exposure of the iridescent and brown side of their wings during erratic flapping flight (Le Roy, Amadori, et al., 2021 [DOI](#); Young, 1971 [DOI](#)).

Overall, the lack of directionality and the gradually changing aspect of *Morpho* iridescent signals do not point at any private iridescent signal. Instead, these visual effects are consistent with the emission of a confusing visual signal that increases the chances of escaping predators. The convergence of visual signals could have been favored because of the indirect advantage that results in a reduction in attacks from predators that have experienced and learned to avoid the escape capabilities of these prey, consistent with the hypothesis of escape mimicry (Pinheiro & Freitas, 2014 [DOI](#); Ruxton et al., 2004 [DOI](#)).

Interestingly, our reflectance measurements showed that males are generally brighter than females in both species, implying that the flashes generated by males during flight could be more conspicuous. Field experiments have highlighted the territorial behavior of males found patrolling along rivers and displaying competitive interactions with other males (Le Roy, Roux, et al., 2021 [DOI](#)). This flight behavior strongly differs from female behavior, which are much less frequently observed in the wild and typically spend more time hidden in the understory (Young, 1973 [DOI](#)). Considering that these ecological niche differences between males and females could result in different predation pressures in contrasted light conditions (open river banks vs. cluttered understory), iridescent signals could have evolved differently in the two sexes, consistent with a similar effect of divergent micro-habitats of males and females in both species (Allen et al., 2011 [DOI](#); Reimchen & Nosil, 2004 [DOI](#)).

Altogether, our results show that natural selection exerted by predators may have favored the evolution of mimetic iridescent color patterns, as well as sexual dimorphism in sympatric *Morpho* species.

Evolution of visual and olfactory cues in mimetic sister-species living in sympatry

Our tetrad and male mate choice experiments in *M. helenor* butterflies clearly show assortative mating based on wing coloration. This result is consistent with the high attraction generated by blue coloration in males, as well as the preference of males for local vs. exotic wing color patterns found in previous field behavioral experiments carried out in Amazonian Peru (Le Roy, Roux, et al., 2021 [↗](#)). Our male mate choice experiments suggest that both the wing patterning and the iridescent properties of the blue patch could have an effect on male preference. Furthermore, visual modeling also suggests that the differences of iridescent coloration between the populations of the same species can be perceived by mates. Coloration and color perception are key in mate recognition in many taxa (marine invertebrates (Baldwin & Johnsen, 2009 [↗](#)), birds (Caro et al., 2021 [↗](#)), mammals (Waitt et al., 2003 [↗](#))) but patterns can also inform mate choice (Houde, 1987 [↗](#); Pérez-Rodríguez et al., 2017 [↗](#)). In jumping spiders, both pattern and coloration are important cues used during mate choice (Zhou et al., 2021 [↗](#)). Similarly, in *Heliconius* butterflies, both pigmentary coloration and wing pattern are used in mate choice, although the preference is hierarchical and coloration is the most important cue (Finkbeiner et al., 2014 [↗](#)). Further experiments are needed to disentangle the relative effects of iridescent coloration and wing pattern in mate preference in *Morpho* butterflies, but our results show that iridescent color patterns can be used as mate recognition cues in *M. helenor*.

Interestingly, intraspecific variations in courtship motivation were detected in our behavioral experiments within the species *M. helenor*. Individuals from the western areas of the Andes displayed high courtship motivation while butterflies from the Amazonian area, where *M. helenor* coexists with *M. achilles*, showed a strong decrease in responses to our visual dummies. The joint convergence of the color pattern and blue iridescence between *M. helenor* and *M. achilles* in Amazonia probably makes visual cues poor species recognition signals: visual modeling even suggests that *M. helenor* cannot perceive the color difference between the two. This is further supported by our male choice experiment showing that although males are attracted by blue iridescent colors and approach each female, they do not preferentially interact with their conspecific female dummies. Selection on visual preference might be relaxed in Amazonian areas where the two sister species are found in sympatry: the shared predator communities may strongly favor convergent evolution of dorsal iridescent color pattern, making visual signals poor species recognition cue.

The convergent evolution of iridescence in sympatric species might thus promote divergent evolution of alternative mating cues. By quantifying chemical compounds found on the genitalia of males and females sampled in Amazonian French Guiana, we did indeed find a strong divergence in the chemical profiles of sympatric *M. helenor* and *M. achilles*, especially in males. High male chemical divergence in sympatric species suggests the evolution of female mate choice based on olfactory discrimination (Mérot et al., 2015 [↗](#)). Although all compounds were not identified, some of those that were have previously been identified as pheromone compounds used during male/female interactions in butterflies. In particular, beta-ocimene was found on the genitalia of males and is known to contribute to mate choice in courtship (Li et al., 2017 [↗](#)) or as an anti-aphrodisiac used by *Heliconius* males to repel other males after mating with a female (Schulz et al., 2008 [↗](#)). Additionally, we found strong divergence between males for heavier chemical compounds (C16-C30). Because cuticular hydrocarbons can be involved in mate discrimination during courtship (Ômura et al., 2020 [↗](#)), diverging non-volatile compounds could also be used as a discriminating cue. Although we did not perform mate choices to test for the discrimination of olfactory cue by females, divergence of odorant cues could be due to reproductive interference

between sympatric species (Bacquet et al., 2015 [↗](#); Dyer et al., 2014 [↗](#)). Altogether, our results therefore strongly suggest that the convergent evolution of iridescent wing pattern between the sympatric *M. helenor* and *M. achilles* in Amazonia, likely promoted by shared predation pressures, have relaxed selection on visually-based preferences and may have favored the evolution of divergent olfactory cues.

Conclusions

Overall, our study shows that local ecological interactions could result in iridescent color patterns and mate preferences evolving separately in closely-related sister species. We find evidence of converging iridescent patterns in sympatry suggesting that predation could play a major role in the evolution of iridescence, by favoring the emergence of evasive mimicry in a similar manner as the Müllerian mimicry of chemically-defended species (Pinheiro et al., 2016 [↗](#)). There are nevertheless striking differences between the two types of mimicry: while a wide diversity of conspicuous color patterns are involved in Müllerian mimicry (Mallet & Gilbert, 1995 [↗](#)), the variation of iridescent patterns involved in evasive mimicry might be more tightly constrained, because iridescence itself directly participates in survival via visual confusion (Murali & Kodandaramaiah, 2020 [↗](#)). Moreover, switches in color pattern is generally considered as a magic trait facilitating speciation in chemically defended species (Merrill et al., 2012 [↗](#)), and Müllerian mimicry frequently occurs among distantly-related species (Puissant et al., 2023 [↗](#)). Here we observe the co-existence of *Morpho* sister-species displaying convergent color patterns, suggesting that speciation can be promoted by the evolution of traits other than visual such as olfactory chemical cues, or by the specialization in other ecological temporal niches (Le Roy, Roux, et al., 2021 [↗](#)).

Altogether, this study shows how convergent trait evolution might lead to mutualistic interactions enhancing species coexistence (Boussens-Dumon & Laurens, 2021 [↗](#)), and alter how selection acts on other traits. It illustrates how the feedback between trait evolution and ecological interactions can fuel speciation and co-existence in sympatry.

Materials and methods

I. Study system

We investigated trait variations in the sister-species *M. achilles* and *M. helenor*, that occur in sympatry in the Amazonian basin (Fig 1 [↗](#)), with two main sampling schemes, focusing on intraspecific and interspecific variations respectively.

M. helenor populations of Ecuador: Pupae from the Ecuadorian populations of *M. h. bristowi* and *M. h. theodorus* were purchased from a breeding farm located in Ecuador (Quinta De Goulaine, <https://quintadegoulaine.com/es/papillons.php> [↗](#)). These two subspecies of *M. helenor* are found in western vs. eastern areas of the Ecuadorian Andes respectively. These commercially-bought pupae were then raised in insectaries at STRI in Gamboa, Panama, between January and March 2023.

Sympatric species of French Guiana: *M. h. helenor* and *M. a. achilles* individuals were wild-caught and raised in insectaries between July and September 2023 in the Amazonian forest of the Kaw Mountains (GPS coordinates: 4.57, -52.21), French Guiana, France.

II. Quantification and characterization of coloration and iridescence of *Morpho* butterfly wings

A. Measures of wing reflectance

We investigated the variations in iridescence in *M. h. bristowi*, *M. h. theodorus* (from Ecuador), *M. h. helenor* and *M. a. achilles* (from French Guiana), using 10 females and 10 males of each. We estimated the iridescence of the blue patches of the dorsal side of the wings by performing a series of reflectance measurements on wings positioned on a flat surface, following a template to ensure that the wing orientation remained the same across all measurements. Reflectance spectra captured at different angles were recorded on the right forewing, at a specific point located in the wing zone defined by the M3 and Cu1 veins according to the nymphalid ground plan (see (Martin & Reed, 2010 [↗](#))). Wing reflectance was measured using a spectrophotometer (AvaSpec-ULS2048CL-EVO- RS 200-1100nm, Avantes) coupled with a deuterium halogen light source (AvaLight-DH-S-BAL, Avantes) for 21 combinations of illumination and observation angles, covering the proximo-distal and anteroposterior planes of the wing (Fig 2 [↗](#).A, see Supporting Information Text 1 for extended methods).

These measurements were repeated 3 times for all individuals to estimate the repeatability of the reflectance measurements at each angle using the R package *rptR* 0.9.22 (Stoffel et al., 2017 [↗](#)). Since no difference was detected between replicates for any of the tested angles (S12 Fig), the mean of these 3 measurements at each angle for each individual was used in the subsequent analyses.

B. Statistical analysis of iridescence

Each illumination/observation position produced a reflectance spectrum whose characteristics (*i.e.* distribution of reflectance) define the optical properties of the illuminated surface. We focused our analyses on the wavelength range [300, 700 nm], relevant for butterfly and bird observers. All reflectance spectra were analyzed using the R package *pavo2* 2.9.0 (Maia et al., 2019 [↗](#)).

a. Extraction and analysis of colorimetric variables

To identify the main optical effect produced by the wing at each angle, we extracted three commonly-used colorimetric variables from each spectrum, using the `summary()` function in *pavo2*: namely Hue (color in the common meaning of the term - measured as the wavelength at the maximum reflectance), Brightness (average proportion of reflected light - measured by the mean reflectance over each spectra) and Chroma (also referred to as saturation, it characterizes the color's purity - measured as the ratio of the reflectance range of each spectra by the brightness). Because the normality and heteroscedasticity assumptions were not met for the distributions of values in these three parameters, we relied on permutation-based ANOVAs (*vegan* 2.6.4 package, (Oksanen et al., 2001 [↗](#))) to test the effect of sex, angle of measurement, locality and species on these three colorimetric variables. To specifically test whether differences in iridescence patterns were observed among species or localities, we estimated the interaction between the effects of *Morpho* populations and of the angle of illumination.

b. Global iridescence analysis

As iridescence broadly refers to any change in reflectance in relation to illumination and observation angle (Doucet & Meadows, 2009 [↗](#)), we characterized the iridescence of each individual using all complete spectra obtained at all angular positions (400 wavelengths x 21 angular positions). We applied a Principal Component Analysis (PCA) to this dataset, where each dot corresponds to the iridescence of an individual wing and thereby showing the similarities and differences in iridescence among groups. The samples from Ecuador and French Guiana were analyzed separately to test for more precise differences in iridescence between allopatric and

sympatric *Morpho* butterfly populations. The coordinates of the 10 first PCs (representing 98% of the explained variance) were then extracted and a PERMANOVA was performed to test for the effect of sex, locality and species on iridescence patterns.

III. Modeling how *Morpho* butterflies and avian predators perceive *M. helenor*

To investigate whether the variations observed among the different spectra could be perceived by different observers, we then used the R package *pavo2* 2.9.0 (Maia et al., 2019) to model both avian and butterfly visual capabilities. We used the avian UV-sensitive vision model implemented in *pavo* (“avg.uv”) as a proxy for predator vision and specifically built a model of *M. helenor* vision, using the spectral sensitivity measured in (Pirih et al., 2022). We built an hexachromatic visual model using the *sensmodel* function, using 345, 446, 495, 508, 570 and 602 nm as the maximum of absorbance in our six simulated cones respectively (S10 Fig). The *vismodel* function was used to compute the quantum catch of each receptor in the *Morpho* vision model under standard daylight illumination for each spectrum. Finally, assuming that the discrimination of colors is limited by photoreceptor noise (Receptor-Noise Limited model, (Vorobyev et al., 2001)), we estimated the distance between colored stimuli as perceived by Morphos using the *bootcoldist* function. We used the default options (*weber* = 0.1, *weber.achro* = 0.1) and set the photoreceptor densities for each cones used in the visual model to 1. The resulting chromatic distances (*dS*) and confidence interval were then used to test how much the visual capacities of *Morpho* butterflies could allow the discrimination of wing reflectance depending on sex, localities and species, at each angle. Similarly, we used the avian UV sensitive visual model implemented in *pavo* to test the discrimination of *Morpho* wing reflectance by predators. We used *dS* ≥ 1 *Just Noticeable Difference* as the threshold of discrimination.

IV. Mate choice experiments

All mate choice experiments took place in cages of similar dimensions (3m x 3m x 2m). All *Morpho* butterflies were fed 30 minutes before the beginning of the experiments taking place between 8am and 12am.

a. Testing for mate discrimination within *M. helenor*

To study the discrimination of mating partners within the species *M. helenor*, tetrad experiments were performed between *M. h. bristowi* and *M. h. theodorus*. For each experiment, one virgin female and one mature male of both *M. h. bristowi* and *M. h. theodorus* were placed within the same cage. Their starting positions in each corner of the cage were randomized. The first mating occurring out of the four possibilities was recorded and the experiment was stopped. Experiments could also be ended when no mating occurred within 5 hours of trial. We carried out N = 33 trials to estimate whether the first mating observed preferentially involved assortative pairs, due to either male and/or female mate preference, of which N=30 trials actually resulted in assortative or disassortative matings. We statistically tested for departure from the random mating expectation using a *chi-square* test of independence, confirmed with a Fisher’s exact test.

Note that *M. achilles* is not commercially bred and wild females are usually all mated, preventing experiments on virgin females in this species, and thus restricting experiments on virgin females in the commercially-bred species *M. helenor*.

b. Identifying specific visual cues used during mate choice by *M. helenor* males

To investigate the visual preference of *M. helenor* males for the dorsal color pattern of females, we performed several choice experiments using dummy butterflies built with actual female wings. To ensure that preferences were not triggered by olfactory cues, all female wings used to build the

artificial dummies were washed with hexane before any experiment. The details of the experimental design and specific visual cues tested are described in **Figure 6** [↗](#).

First, we tested male preference for the visual appearance of *M. h. bristowi* and *M. h. theodorus* females (Experiment 1). Since *M. h. bristowi* and *M. h. theodorus* females differ both in terms of color patterns (width of blue band) and iridescence, we then conducted experiments to isolate these two components.

We specifically tested the effect of the proportion of blue to black areas (Experiment 2) by presenting males *M. h. bristowi* to female dummies harboring *M. h. bristowi* wings, except one was modified with black ink to resemble the narrow blue band on *M. h. theodorus* female wings.

We finally tested the effect of the blue iridescent band within the same black and blue pattern (Experiment 3) by presenting *M. h. theodorus* males to a *M. h. theodorus* female with a narrow blue band, and to the modified *M. h. bristowi* female dummy with a narrow band. The wing pattern modifications were performed using a black permanent marker.

To ensure that ink odor would not bias the behavior of *Morpho* males, all dummies used in experiments 2 and 3 contained traces of black marker (either on the modified bands or on the ventral side of the wings of natural females).

In all experiments, two female dummies were hanging from the ceiling of the cage, 1.5 m apart, and presented to a live male. The male was released in the experimental cage in front of the two female dummies and was free to interact with them for 10 minutes. The positions of the female models were swapped 5 minutes into each trial to avoid potential position-related bias. Two types of male behaviors were recorded. Each time a male flew within a 40 cm radius from a female model, an “approach” was recorded. Each time a male touched a female dummy, a “touch” was recorded. The number of “approaches” and “touches” toward each female dummy per trial was used to determine male preference. Each male performed this experiment 3 times on 3 different days to account for potential individual variations in mate preference.

c. Testing whether visual cues can act as a pre-zygotic barrier between *M. h. helenor* and *M. a. achilles*

In order to investigate the importance of color vision in species recognition, we then performed a male choice experiment with *M. h. helenor* and *M. a. achilles* (experiment 4 in **Figure 6** [↗](#)) in French Guiana, using natural wings of females *M. h. helenor* and *M. a. achilles* as artificial butterflies following the exact same protocol.

d. Statistical analyses

Comparing male mating motivation

Because we noticed behavioral differences between the species and subspecies of *Morpho* during the course of the experiments 1-4, we attempted to account for variations in male mating motivation. The approach and touch rates of males, *i.e.* their willingness to approach and touch (respectively) any female dummies during each trial, was defined as the proportion of trials where the males approached and touched (respectively) either female dummies at least once. The approach rate and touch rate of *M. h. bristowi*, *M. h. theodorus*, *M. h. helenor* and *M. a. achilles* males were calculated, and a binomial generalized linear model (GLM) was used to test whether the approach or touch rate was significantly different between males from different populations.

All statistical analyses were performed with the R package *glmmTMB* (Brooks et al., 2017 [↗](#)). The normality, homogeneity of variance and overdispersion tests for the GLMs were performed using the R package *DHARMA* (Hartig, 2022 [↗](#)). We controlled for potential individual variations by

Experiments (Location)	Dummy female 1	Dummy female 2	Live males	Test
1 (STRI)	<i>M. h. theodorus</i> 	<i>M. h. bristowi</i> 	<i>M. h. theodorus</i> (n = 33)	Assortative preference within species
			<i>M. h. bristowi</i> (n = 28)	
2 (STRI)	<i>M. h. bristowi</i> 	<i>M. h. bristowi</i> with thinner blue band 	<i>M. h. bristowi</i> (n = 28)	Discrimination on the pattern
3 (STRI)	<i>M. h. theodorus</i> 	<i>M. h. bristowi</i> with thinner blue band 	<i>M. h. theodorus</i> (n = 33)	Discrimination on the iridescent blue
4 (French Guiana)	<i>M. h. helenor</i> 	<i>M. a. achilles</i> 	<i>M. h. helenor</i> (n = 28)	Assortative preference between species
			<i>M. a. achilles</i> (n=37)	

Figure 6

Experimental design of the male choice experiments

performed between the Morphos originating from Ecuadorian populations of *M. helenor* (eastern population: *M. h. theodorus* and western population: *M. h. bristowi*, experiments 1 to 3) and from French Guiana (*M. h. helenor* and *M. a. achilles*, experiment 4), along with the associated preferences and visual cues tested. Note that the same males were used for experiments 1, 2 and 3.

using the male ID and trial number as a random effect in all the tests.

Visually-based male mate preferences

We then tested for male preferential attraction for different dummy females. The mean proportion of approaches and touches toward each female dummy was calculated over the four types of trials performed by each male. Non-parametric Wilcoxon tests were used to test whether the mean probability of approaching or touching a dummy 1/dummy 2 was significantly different from the 0.5 proportion expected under random mating. A significant male preference towards either dummy was detected when the mean probability was significantly above 0.5.

V. Pheromone analysis for sympatric species

a. Chemical analyses

We investigated variations in chemical compounds between the two sympatric species *M. helenor* and *M. achilles* sampled in French Guiana. The male and female genitalia of wild caught *M. h. helenor* and *M. a. achilles* individuals (from 9 up to 12 per sex per species) were dissected and extracted in 20 μ L of dichloromethane, containing 5 μ L of octadecane as an internal standard. Two sets of chromatography were performed in order to extract either the light compounds (all volatile carbonate compounds below C16) or the heavier ones (all carbonate compounds above C16) found on the genitalia of those Morphos.

The chemical compounds of those samples were analyzed by gas chromatography and time-of-flight mass spectrophotometry (GC-ToF-MS) using a 8890 GC system (Agilent, Santa Clara, USA) coupled to a PEGASUS BT mass spectrophotometer (LECO, Saint Joseph, USA). The connected capillary column was a Rxi-5ms (30 m x 0.25 mm, df = 0.25 μ m, Restek, Bellefonte USA) and helium was used as carrier gas at 1 mL min⁻¹ constant flow rate. 1 μ L of each sample was injected in a GC split/splitless injector at 300 °C. Two temperature programs were used: (i) the analysis of the C8-C16 compounds started at 40 °C maintained for 1 min, then increased at 210 °C at 6 °C min⁻¹ rate, finally increased to 340 °C at a 10 °C min⁻¹, and (ii) the analysis of the C16-C30 compounds started at 70 °C maintained for 1 min, then increased to 150 °C at 30 °C min⁻¹ rate and finally increased to 340 °C at a 6 °C min⁻¹ rate. Mass spectra were recorded with a 70 eV electron impact energy. Mixtures of alkanes C8 to C19 and C16 to C44 (ASTM® D5442 C16-C44 Qualitative Retention Time Mix, Supelco, Bellefonte, USA) were injected under the same conditions to be used as external standards.

The chromatograms were converted in netCDF format and operated using *MZmine 2.43* for compound detection and area integration. To this aim, chromatograms were cropped before 4 min and after 27 min, and baseline-corrected using the Rolling ball algorithm (Kneen & Annegarn, 1996 [\[4\]](#)). Mass detection was performed using the centroid mass detector, and chromatograms of each mass were constructed using the ADAP algorithm (Myers et al., 2017 [\[4\]](#); Pluskal et al., 2010 [\[4\]](#)). Chromatograms were deconvoluted into individual peaks using the local minimum search algorithm and aligned on a master peak list with the RANSAC algorithm. Finally, the Peak Finder gap filling algorithm was used to search for missing peaks, and a matrix compiling the area under each peak detected for each butterfly sample was extracted. This matrix was purged from siloxane like compounds as well as compounds present in less than 10% of the samples. The areas of the blanks were subtracted to the samples, and the final data were used for statistical analysis. Mass spectra were then visualized using the ChromaTOF® software, compared to MZmine-selected ones, in order to manually annotate the chemical compounds. Annotation were based on scan comparison with the NIST2017 database, and linear retention indexes (LRI) calculated from the alkane mixtures with free database (<https://pubchem.ncbi.nlm.nih.gov/> [\[4\]](#)) and literature (Adams, 2017).

b. Statistical analyses of the chemical compounds

In order to quantify variations in chemical compounds detected in the different sample, we performed a nonmetric multidimensional scaling (NMDS) ordination using the Bray-Curtis dissimilarity calculation. By considering the amount of each compound found in our samples as a variable, we performed a permutation MANOVA with the `adonis2()` function found in the *vegan 2* package on R. We tested whether the sex or the species had a significant effect on the chemical bouquet produced by their genitalia.

We also performed an Indicator Value Analysis to identify the chemical compounds significantly associated to our different *Morpho* samples with the *mutlipatt* function from the *indicspecies* package in R using the IndVal index from (Dufrêne & Legendre, 1997 [↗](#)).

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Additional files

Supplementary files [↗](#)

References

- Allen C. E., Zwaan B. J., Brakefield P. M (2011) **Evolution of Sexual Dimorphism in the Lepidoptera** *Annual Review of Entomology* **56** <https://doi.org/10.1146/annurev-ento-120709-144828>Google Scholar
- Bacquet P. M. B., Brattström O., Wang H.-L., Allen C. E., Löfstedt C., Brakefield P. M., Nieberding C. M (2015) **Selection on male sex pheromone composition contributes to butterfly reproductive isolation** *Proceedings of the Royal Society B: Biological Sciences* **282**:20142734 <https://doi.org/10.1098/rspb.2014.2734>Google Scholar
- Baldwin J., Johnsen S (2009) **The importance of color in mate choice of the blue crab Callinectes sapidus** *Journal of Experimental Biology* **212**:3762–3768 <https://doi.org/10.1242/jeb.028027>Google Scholar
- Berlocher S. H., Feder J. L (2002) **Sympatric Speciation in Phytophagous Insects : Moving Beyond Controversy?** *Annual Review of Entomology* **47** <https://doi.org/10.1146/annurev.ento.47.091201.145312>Google Scholar
- Blandin P., Purser B (2013) **Evolution and diversification of Neotropical butterflies : Insights from the biogeography and phylogeny of the genus Morpho Fabricius, 1807 (Nymphalidae : Morphinae), with a review of the geodynamics of South AAmerica** *Tropical Lepidoptera Research* Google Scholar
- Boussens-Dumon G., Llaurens V (2021) **Sex, competition and mimicry : An eco-evolutionary model reveals unexpected impacts of ecological interactions on the evolution of phenotypes in sympatry** *Oikos* **130**:2028–2039 <https://doi.org/10.1111/oik.08139>Google Scholar
- Brooks M. E., Kristensen K., van Benthem K. J., Magnusson A., Berg C. W., Nielsen A., Skaug H. J., Machler M., Bolker B. M. (2017) **glmmTMB balances speed and flexibility among packages for zero-inflated generalized linear mixed modeling** *The R Journal* **9**:378–400 <https://doi.org/10.32614/RJ-2017-066>Google Scholar
- Butlin R. K., Smadja C. M (2018) **Coupling, Reinforcement, and Speciation** *The American Naturalist* **191**:155–172 <https://doi.org/10.1086/695136>Google Scholar
- Caro S. P., Pierre L., Bergès M., Bakker R., Doutrelant C., Bonadonna F (2021) **Mutual mate preferences and assortative mating in relation to a carotenoid-based color trait in blue tits** *Behavioral Ecology* **32**:1171–1182 <https://doi.org/10.1093/beheco/arab080>Google Scholar
- Chazot N., Blandin P., Debat V., Elias M., Condamine F. L (2021) **Punctuational ecological changes rather than global factors drive species diversification and the evolution of wing phenotypes in Morpho butterflies** *Journal of Evolutionary Biology* **34**:1592–1607 <https://doi.org/10.1111/jeb.13921>Google Scholar
- Chazot N., Willmott K. R., Santacruz Endara P. G., Toporov A., Hill R. I., Jiggins C. D., Elias M (2014) **Mutualistic Mimicry and Filtering by Altitude Shape the Structure of Andean Butterfly Communities** *The American Naturalist* **183**:26–39 <https://doi.org/10.1086/674100>Google Scholar

- Chouteau M., Arias M., Joron M (2016) **Warning signals are under positive frequency-dependent selection in nature** *Proceedings of the National Academy of Sciences* **113**:2164–2169 <https://doi.org/10.1073/pnas.1519216113>Google Scholar
- Costanzo K., Monteiro A (2007) **The use of chemical and visual cues in female choice in the butterfly *Bicyclus anynana*** *Proceedings of the Royal Society B: Biological Sciences* **274**:845–851 <https://doi.org/10.1098/rspb.2006.3729>Google Scholar
- Doucet S. M., Meadows M. G (2009) **Iridescence : A functional perspective** *Journal of The Royal Society Interface* **6** <https://doi.org/10.1098/rsif.2008.0395.focus>Google Scholar
- Dufrêne M., Legendre P (1997) **Species Assemblages and Indicator Species:the Need for a Flexible Asymmetrical Approach** *Ecological Monographs* **67**:345–366 [https://doi.org/10.1890/0012-9615\(1997\)067\[0345:SAAIST\]2.0.CO;2](https://doi.org/10.1890/0012-9615(1997)067[0345:SAAIST]2.0.CO;2)Google Scholar
- Dyer K. A., White B. E., Sztepanacz J. L., Bewick E. R., Rundle H. D (2014) **Reproductive character displacement of epicuticular compounds and their contribution to mate choice in *Drosophila subquinaria* and *Drosophila recens*** *Evolution* **68**:1163–1175 <https://doi.org/10.1111/evo.12335>Google Scholar
- Ellers J., Boggs C. L (2003) **The evolution of wing color : male mate choice opposes adaptive wing color divergence in colias butterflies** *Evolution* **57**:1100–1106 <https://doi.org/10.1111/j.0014-3820.2003.tb00319.x>Google Scholar
- Endler J. A (1992) **Signals, Signal Conditions, and the Direction of Evolution** *The American Naturalist* **139**:S125–S153 <https://doi.org/10.1086/285308>Google Scholar
- Estrada C., Jiggins C. D (2008) **Interspecific sexual attraction because of convergence in warning colouration : Is there a conflict between natural and sexual selection in mimetic species?** *Journal of Evolutionary Biology* **21**:749–760 <https://doi.org/10.1111/j.1420-9101.2008.01517.x>Google Scholar
- Finkbeiner S. D., Briscoe A. D., Reed R. D (2014) **Warning signals are seductive : Relative contributions of color and pattern to predator avoidance and mate attraction in *Heliconius* butterflies: BUTTERFLY VISUAL SIGNALS** *Evolution* **68**:3410–3420 <https://doi.org/10.1111/evo.12524>Google Scholar
- Giraldo M. A., Yoshioka S., Liu C., Stavenga D. G (2016) **Coloration mechanisms and phylogeny of *Morpho* butterflies** *Journal of Experimental Biology* **219**:3936–3944 <https://doi.org/10.1242/jeb.148726>Google Scholar
- González-Rojas M. F., Darragh K., Robles J., Linares M., Schulz S., McMillan W. O., Jiggins C. D., Pardo-Díaz C., Salazar C (2020) **Chemical signals act as the main reproductive barrier between sister and mimetic *Heliconius* butterflies** *Proceedings of the Royal Society B: Biological Sciences* **287**:20200587 <https://doi.org/10.1098/rspb.2020.0587>Google Scholar
- Gröning J., Hochkirch A (2008) **Reproductive Interference Between Animal Species** *The Quarterly Review of Biology* **83**:257–282 <https://doi.org/10.1086/590510>Google Scholar
- Gruson H., Andraud C., Daney de Marcillac W., Berthier S., Elias M., Gomez D. (2018) **Quantitative characterization of iridescent colours in biological studies : A novel method**

using optical theory *Interface Focus* **9**:20180049 <https://doi.org/10.1098/rsfs.2018.0049>Google Scholar

Hartig F. (2022) **DHARMA : Residual Diagnostics for Hierarchical (Multi-Level / Mixed) Regression Models** <https://github.com/florianhartig/dharma>

Houde A. E (1987) **Mate Choice Based Upon Naturally Occurring Color-Pattern Variation in a Guppy Population** *Evolution* **41**:1 <https://doi.org/10.1111/j.1558-5646.1987.tb05766.x>Google Scholar

Joron M., Wynne I. R., Lamas G., Mallet J (1999) **Variable Selection and the Coexistence of Multiple mimetic forms of the Butterfly *Heliconius numata*** *Evolutionary Ecology* **13**:721–754 <https://doi.org/10.1023/A:1010875213123>Google Scholar

Kneen M. A., Annegarn H. J (1996) **Algorithm for fitting XRF, SEM and PIXE X-ray spectra backgrounds** *Nuclear Instruments and Methods in Physics Research Section B: Beam Interactions with Materials and Atoms* **109** [https://doi.org/10.1016/0168-583X\(95\)00908-6](https://doi.org/10.1016/0168-583X(95)00908-6)Google Scholar

Le Roy C., Amadori D., Charberet S., Windt J., Muijres F. T., Llaurens V., Debat V. (2021) **Adaptive evolution of flight in Morpho butterflies** *Science* **374**:1158 [Google Scholar](#)

Le Roy C., Roux C., Authier E., Parrinello H., Bastide H., Debat V., Llaurens V. (2021) **Convergent morphology and divergent phenology promote the coexistence of Morpho butterfly species** *Nature communications* **12**:7248 [Google Scholar](#)

Li C., Wang H., Chen X., Yao J., Shi L., Zhou C (2017) **Role of visual and olfactory cues in sex recognition in butterfly *Cethosia cyane cyane*** *Scientific Reports* **7**:5033 <https://doi.org/10.1038/s41598-017-04721-6>Google Scholar

Llaurens V., Joron M., Théry M (2014) **Cryptic differences in colour among Müllerian mimics : How can the visual capacities of predators and prey shape the evolution of wing colours?** *Journal of Evolutionary Biology* **27**:531–540 <https://doi.org/10.1111/jeb.12317>Google Scholar

Llaurens V., Le Poul Y., Puissant A., Blandin P., Debat V. (2021) **Convergence in sympatry : Evolution of blue-banded wing pattern in Morpho butterflies** *Journal of Evolutionary Biology* **34**:284–295 <https://doi.org/10.1111/jeb.13726>Google Scholar

Maan M. E., Seehausen O (2011) **Ecology, sexual selection and speciation** *Ecology Letters* **14**:591–602 <https://doi.org/10.1111/j.1461-0248.2011.01606.x>Google Scholar

Maia R., Gruson H., Endler J. A., White T. E (2019) **2 : New tools for the spectral and spatial analysis of colour in r** *Methods in Ecology and Evolution* **10**:1097–1107 <https://doi.org/10.1111/2041-210X.13174>Google Scholar

Maisonneuve L., Smadi C., Llaurens V (2024) **Which cues are sexy? The evolution of mate preference in sympatric species reveals the contrasted effect of adaptation and reproductive interference** . *Evolution Letters* **8**:283–294 <https://doi.org/10.1093/evlett/qrad058>Google Scholar

- Mallet J., Gilbert L. E. (1995) **Why are there so many mimicry rings? Correlations between habitat, behaviour and mimicry in Heliconius butterflies** *Biological Journal of the Linnean Society* **55**:159–180 <https://doi.org/10.1111/j.1095-8312.1995.tb01057.x>Google Scholar
- Martin A., Reed R. D (2010) **Wingless and aristaless2 Define a Developmental Ground Plan for Moth and Butterfly Wing Pattern Evolution** *Molecular Biology and Evolution* **27**:2864–2878 <https://doi.org/10.1093/molbev/msq173>Google Scholar
- Mérot C., Frérot B., Leppik E., Joron M (2015) **Beyond magic traits : Multimodal mating cues in Heliconius butterflies** *Evolution* **69**:2891–2904 <https://doi.org/10.1111/evo.12789>Google Scholar
- Merrill R. M., Chia A., Nadeau N. J (2014) **Divergent warning patterns contribute to assortative mating between incipient Heliconius species** *Ecology and Evolution* **4**:911–917 <https://doi.org/10.1002/ece3.996>Google Scholar
- Merrill R. M., Wallbank R. W. R., Bull V., Salazar P. C. A., Mallet J., Stevens M., Jiggins C. D (2012) **Disruptive ecological selection on a mating cue** *Proceedings of the Royal Society B: Biological Sciences* **279**:4907–4913 <https://doi.org/10.1098/rspb.2012.1968>Google Scholar
- Murali G., Kodandaramaiah U (2020) **Size and unpredictable movement together affect the effectiveness of dynamic flash coloration** *Animal Behaviour* **162**:87–93 <https://doi.org/10.1016/j.anbehav.2020.02.002>Google Scholar
- Muschick M., Indermaur A., Salzburger W (2012) **Convergent Evolution within an Adaptive Radiation of Cichlid Fishes** *Current Biology* **22**:2362–2368 <https://doi.org/10.1016/j.cub.2012.10.048>Google Scholar
- Myers O. D., Sumner S. J., Li S., Barnes S., Du X (2017) **One Step Forward for Reducing False Positive and False Negative Compound Identifications from Mass Spectrometry Metabolomics Data : New Algorithms for Constructing Extracted Ion Chromatograms and Detecting Chromatographic Peaks** *Analytical Chemistry* **89**:8696–8703 <https://doi.org/10.1021/acs.analchem.7b00947>Google Scholar
- Oksanen J., Simpson G. L., Blanchet F. G., Kindt R., Legendre P., Minchin P. R., O'Hara R. B., Solymos P., Stevens M. H. H., Szoecs E., Wagner H., Barbour M., Bedward M., Bolker B., Borcard D., Carvalho G., Chirico M., De Caceres M., Durand S., ..., Weedon J (2001) **vegan : Community Ecology Package** *CRAN* 2.6–10) <https://doi.org/10.32614/CRAN.package.vegan>
- Oliver J. C., Robertson K. A., Monteiro A (2009) **Accommodating natural and sexual selection in butterfly wing pattern evolution** *Proceedings of the Royal Society B: Biological Sciences* **276**:2369–2375 <https://doi.org/10.1098/rspb.2009.0182>Google Scholar
- Ômura H., Noguchi T., Ohta S (2020) **The male swallowtail butterfly, Papilio polytes, uses cuticular hydrocarbons for mate discrimination** *Animal Behaviour* **170**:133–145 <https://doi.org/10.1016/j.anbehav.2020.10.016>Google Scholar
- Páez E., Valkonen J. K., Willmott K. R., Matos-Maraví P., Elias M., Mappes J (2021) **Hard to catch : Experimental evidence supports evasive mimicry** *Proceedings of the Royal Society B: Biological Sciences* **288**:20203052 <https://doi.org/10.1098/rspb.2020.3052>Google Scholar
- Pérez-Rodríguez L., Jovani R., Stevens M (2017) **Shape matters : Animal colour patterns as signals of individual quality** *Proceedings of the Royal Society B: Biological Sciences* **284**:20162446 <https://doi.org/10.1098/rspb.2016.2446>Google Scholar

- Pfennig K. S., Pfennig D. W (2009) **Character Displacement : Ecological and Reproductive Responses to a Common Evolutionary Problem** *The Quarterly Review of Biology* **84**:253–276 <https://doi.org/10.1086/605079>Google Scholar
- Pinheiro C. E. G., Freitas A. V. L (2014) **Some Possible Cases of Escape Mimicry in Neotropical Butterflies** *Neotropical Entomology* **43**:393–398 <https://doi.org/10.1007/s13744-014-0240-y>Google Scholar
- Pinheiro C. E. G., Freitas A. V. L., Campos V. C., DeVries P. J., Penz C. M (2016) **Both Palatable and Unpalatable Butterflies Use Bright Colors to Signal Difficulty of Capture to Predators** *Neotropical Entomology* **45**:107–113 <https://doi.org/10.1007/s13744-015-0359-5>Google Scholar
- Pirih P., Ilić M., Meglič A., Belušič G (2022) **Opponent processing in the retinal mosaic of nymphalid butterflies** *Philosophical Transactions of the Royal Society B: Biological Sciences* **377**:20210275 <https://doi.org/10.1098/rstb.2021.0275>Google Scholar
- Pluskal T., Castillo S., Villar-Briones A., Orešič M (2010) **MZmine 2 : Modular framework for processing, visualizing, and analyzing mass spectrometry-based molecular profile data** *BMC Bioinformatics* **11**:395 <https://doi.org/10.1186/1471-2105-11-395>Google Scholar
- Puissant A., Chotard A., Condamine F. L., Llaurens V (2023) **Convergence in sympatric swallowtail butterflies reveals ecological interactions as a key driver of worldwide trait diversification** *Proceedings of the National Academy of Sciences* **120**:e2303060120 <https://doi.org/10.1073/pnas.2303060120>Google Scholar
- Reimchen T. E., Nosil P (2004) **Variable Predation Regimes Predict the Evolution of Sexual Dimorphism in a Population of Threespine Stickleback** *Evolution* **58**:1274–1281 <https://doi.org/10.1111/j.0014-3820.2004.tb01706.x>Google Scholar
- Reynolds R. G., Fitzpatrick B. M (2007) **Assortative mating in poison-dart frogs based on an ecologically important trait** *Evolution* **61**:2253–2259 <https://doi.org/10.1111/j.1558-5646.2007.00174.x>Google Scholar
- Ruxton G. D., Speed M., Sherratt T. N (2004) **Evasive mimicry : When (if ever) could mimicry based on difficulty of capture evolve?** *Proceedings of the Royal Society of London. Series B: Biological Sciences* **271**:2135–2142 <https://doi.org/10.1098/rspb.2004.2816>Google Scholar
- Schulz S., Estrada C., Yildizhan S., Boppré M., Gilbert L. E (2008) **An Antiaphrodisiac in Heliconius melpomene Butterflies** *Journal of Chemical Ecology* **34**:82–93 <https://doi.org/10.1007/s10886-007-9393-z>Google Scholar
- Seehausen O., Terai Y., Magalhaes I. S., Carleton K. L., Mrosso H. D. J., Miyagi R., van der Sluijs I., Schneider M. V., Maan M. E., Tachida H., Imai H., Okada N. (2008) **Speciation through sensory drive in cichlid fish** *Nature* **455**:620–626 <https://doi.org/10.1038/nature07285>Google Scholar
- Servedio M. R (2004) **The evolution of premating isolation : local adaptation and natural and sexual selection against hybrids** *Evolution* **58**:913–924 <https://doi.org/10.1111/j.0014-3820.2004.tb00425.x>Google Scholar
- Servedio M. R., Boughman J. W (2017) **The Role of Sexual Selection in Local Adaptation and Speciation** *Annual Review of Ecology, Evolution, and Systematics* **48** <https://doi.org/10.1146/annurev-ecolsys-110316-022905>Google Scholar

- Siddique R. H., Diewald S., Leuthold J., Hölscher H (2013) **Theoretical and experimental analysis of the structural pattern responsible for the iridescence of Morpho butterflies** *Optics Express* **21**:14351 <https://doi.org/10.1364/OE.21.014351> Google Scholar
- Simpson R. K., McGraw K. J (2019) **Interspecific Covariation in Courtship Displays, Iridescent Plumage, Solar Orientation, and Their Interactions in Hummingbirds** *The American Naturalist* **194**:441–454 <https://doi.org/10.1086/704774> Google Scholar
- Soni S. P., Apte V., Joshi P., Cyriac V. P (2024) **Barking up the wrong frog : Global prevalence of misdirected amplexus in anuran amphibians. Biological Journal of the Linnean Society** *blae* **062** <https://doi.org/10.1093/biolinnean/blae062> Google Scholar
- Stoffel M. A., Nakagawa S., Schielzeth H (2017) **rptR : Repeatability estimation and variance decomposition by generalized linear mixed-effects models** *Methods in Ecology and Evolution* **8**:1639–1644 <https://doi.org/10.1111/2041-210X.12797> Google Scholar
- Stuckert A. M. M., Venegas P. J., Summers K (2014) **Experimental evidence for predator learning and Müllerian mimicry in Peruvian poison frogs (Ranitomeya, Dendrobatidae)** *Evolutionary Ecology* **28**:413–426 <https://doi.org/10.1007/s10682-013-9685-4> Google Scholar
- van Doorn G. S., Edelaar P., Weissing F. J. (2009) **On the Origin of Species by Natural and Sexual Selection** *Science* **326**:1704–1707 <https://doi.org/10.1126/science.1181661> Google Scholar
- Vieira-Silva A., Evora G. B., Freitas A. V. L., Oliveira P. S (2024) **The Relevance of Flash Coloration Against Avian Predation in a Morpho Butterfly : A Field Experiment in a Tropical Rainforest** *Ethology* **130**:e13517 <https://doi.org/10.1111/eth.13517> Google Scholar
- Vorobyev M., Brandt R., Peitsch D., Laughlin S. B., Menzel R (2001) **Colour thresholds and receptor noise : Behaviour and physiology compared** *Vision Research* **41**:639–653 [https://doi.org/10.1016/S0042-6989\(00\)00288-1](https://doi.org/10.1016/S0042-6989(00)00288-1) Google Scholar
- Waitt C., Little A. C., Wolfensohn S., Honess P., Brown A. P., Buchanan-Smith H. M., Perrett D. I (2003) **Evidence from rhesus macaques suggests that male coloration plays a role in female primate mate choice** *Proceedings of the Royal Society of London. Series B: Biological Sciences* **270**:S144–S146 <https://doi.org/10.1098/rsbl.2003.0065> Google Scholar
- White T. E., Zeil J., Kemp D. J (2015) **Signal design and courtship presentation coincide for highly biased delivery of an iridescent butterfly mating signal** *Evolution* **69**:14–25 <https://doi.org/10.1111/evo.12551> Google Scholar
- Young A. M (1971) **Wing coloration and reflectance in Morpho butterflies as related to reproductive behavior and escape from avian predators** *Oecologia* **7**:209–222 <https://doi.org/10.1007/BF00345212> Google Scholar
- Young A. M (1973) **Studies on comparative ecology and ethology in adult populations of several species of Morpho butterflies (Lepidoptera : Morphidae)** *Studies on Neotropical Fauna* **8**:17–50 <https://doi.org/10.1080/01650527309360452> Google Scholar
- Zhou W., Yu L., Kwek B. Z. W., Jin G., Zeng H., Li D (2021) **Sexual selection on jumping spider color pattern : Investigation with a new quantitative approach** *Behavioral Ecology* **32**:695–706 <https://doi.org/10.1093/beheco/arab008> Google Scholar

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Reviewer #1 (Public review):

Summary:

In this study, Ledamoisel et al. examined the evolution of visual and chemical signals in closely related *Morpho* butterfly species to understand their role in species coexistence. Using an integrative, state-of-the-art approach combining spectrophotometry, visual modeling, and behavioral mate choice experiments, they quantified differences in wing iridescence and assessed its influence on mate preference in allopatry and sympatry. They also performed chemical analyses to determine whether sympatric species exhibit divergent chemical cues that may facilitate species recognition and mate discrimination. The authors found iridescent coloration to be similar in sympatric *Morpho* species. Furthermore, male mate choice experiments revealed that in sympatry, males fail to discriminate conspecific females based on coloration, reinforcing the idea that visual signal convergence is primarily driven by predation pressure. In contrast, the divergence of chemical signals among sympatric species suggests their potential role in facilitating species recognition and mate discrimination. The authors conclude that interactions between ecological pressures and signal evolution may shape species coexistence.

Strengths:

The study is well-designed and integrates multiple methodological approaches to provide a thorough assessment of signal evolution in the studied species. I appreciate the authors' careful consideration of multiple selective pressures and their combined influence on signal divergence and convergence. Additionally, the inclusion of both visual and chemical signals adds an interesting and valuable dimension to the study, enhancing its importance. Beyond butterflies, this research broadens our understanding of multimodal communication and signal evolution in the context of species coexistence.

Weaknesses:

(1) The broader significance of the findings needs to be better articulated. While the authors emphasize that comparing adaptive traits in sympatry and allopatry provides insights into selective processes shaping reproductive isolation and coexistence, it is unclear what key conceptual or theoretical questions are being addressed. Are these patterns expected under certain evolutionary scenarios? Have they been empirically demonstrated in other systems? The authors should explicitly state the overarching research question, incorporate some predictions, and better contextualize their findings within the existing literature. If the results challenge or support previous work, that should be highlighted to strengthen the study's importance in a broader context.

(2) The motivation for studying visual signals and mate choice in allopatric populations (i.e., at the intraspecific level) is not well articulated, leaving their role in the broader narrative unclear. In particular, the rationale behind experiments 1, 2, and 3 is not well defined, as the authors have not made a strong case for the need for these intraspecific comparisons in the

introduction. This issue is further compounded by the authors' primary focus on signal evolution in sympatry throughout both the results and the discussion. For instance, the divergence of iridescence in allopatry is a potentially interesting result. But the authors have not discussed its implications.

Overall, given that the primary conclusions are based on results and analyses in sympatry, the role of allopatric populations in shaping these conclusions needs to be better integrated and justified. Without a stronger link between the comparative framework and the study's key takeaways, the use of allopatric populations feels somewhat peripheral rather than central to the study's aim. Since the primary conclusions remain valid even without the allopatric comparisons, their inclusion requires a clearer rationale.

(3) While the authors demonstrate that iridescence is indistinguishable to predators in sympatry, they overstate the role of predation in driving convergence. The present study does not experimentally demonstrate that iridescence in this species has a confusion effect or contributes to evasive mimicry. Alternatively, convergence could result from other selective forces, such as signal efficacy due to environmental conditions, rather than being solely driven by predation.

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Reviewer #2 (Public review):

This study presents an investigation of the visual and chemical properties and mating behaviour in *Morpho* butterflies, aimed at addressing the nature of divergence between closely related species in sympatry. The study species consists of three subspecies of *Morpho helenor* (*bristowi*, *theodorus*, and *helenor*), and the conspecific *Morpho achilles achilles*. The authors postulate that whereas the iridescent blue signals of all (sub)species should function as a predator reduction signal (similar to aposematism) and therefore exhibit convergence, the same signals should indicate divergence if used as a mating signal, particularly in sympatric populations. They also assess chemical profiles among the species to assess the potential utility of scent in mediating species/sex discrimination.

The authors first used reflectance spectrometry to calculate hue, brightness, and chroma, plus two measures of "iridescence" (perhaps better phrased as angular dependence) in each (sub)species. This indicated the ubiquitous presence of sexual dimorphism in brightness (males brighter), which also appears to be the case for iridescence (Figure 3A-B). Analysis of these data also indicated that whereas there is evidence for divergence among subspecies in allopatry, the same evidence is lacking for species in sympatry ($P = 0.084$). This was supported further by visual modelling, which showed that both conspecifics and birds should be (theoretically) capable of perceiving the colour difference among allopatric populations of *M. helenor*, whereas the same is not true for the sympatric species.

The authors then conducted mate choice trials, first using live individuals and second using female dummies. The live experiments indicated the presence of assortative mating among the two subspecies of *M. helenor* (*bristowi* and *theodorus*). The dummy presentations indicated (a) *bristowi* males prefer conspecific wings, whereas *theodorus* have no preference, (b) *bristowi* males prefer the con(sub)specific colour pattern, (c) *theodorus* prefer the con(sub)specific iridescence when the pattern is manipulated to be similar among female dummies. A fourth experiment, using sympatric *M. achilles* and *M. helenor*, indicated no preference for conspecific female dummies. Finally, chemical analysis indicated substantial differences between these two species in putative pheromone compounds, and especially so in the males.

The authors conclude that the similarity of iridescence among species in sympatry is suggestive of convergence upon a common anti-predation signal. Despite some behavioural evidence in favour of colour (iridescence)-based mate discrimination, chemical differences between *Achilles* and *Helenor* are posed as more likely to function for species isolation than visual differences.

Overall, I enjoyed reading this manuscript, which presents a valiant attempt at studying visual, chemical and behavioural divergence in this iconic group of butterflies.

Major comments

My only major comment concerns the authors' favoured explanation for aposematism (or evasive mimicry) for convergence among species, which is based upon the you-can't-catch-me hypothesis first presented by Young 1971. Although there is supporting work showing that iridescent-like stimuli are more difficult to precisely localize by a range of viewers, most of the evidence as applied to the *Morpho* system is circumstantial, and I'm not certain that there is widespread acceptance of this hypothesis. Given that the present study deals with closely-related (sub)species, one alternative explanation - a "null" hypothesis of sorts - is for a lack of divergence (from a common starting point) as opposed to evolutionary convergence *per se*. In other words, two subspecies are likely to retain ancestral character states unless there is selection that causes them to diverge. I feel that the manuscript would benefit from a discussion of this alternative, if not others. Signalling to predators could very well be involved in constraining the extent of convergence, but this seems a little premature to state as an up-front conclusion of this work. There is also the result of a *dorsal* wing manipulation by Vieira-Silva et al. 2024 (<https://doi.org/10.1111/eth.13517>), which seems difficult to reconcile in light of this explanation. Whereas this paper is cited by the authors, a more nuanced discussion of their experimental results would seem appropriate here.

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Reviewer #3 (Public review):

The authors investigated differences in iridescence wing colouration of allopatric (geographically separated) and sympatric (coexisting) *Morpho* butterfly (sub)species. Their aim was to assess if iridescence wing colouration of *Morpho* (sub)species converged or diverged depending on coexistence and if iridescence wing colouration was involved in mating behaviour and reproductive isolation. The authors hypothesize that iridescence wing colouration of different (sub)species should converge in sympatry and diverge in allopatry. In sympatry, iridescence wing colouration can act as an effective antipredator defence with shared benefits if multiple (sub)species share the same colouration. However, shared wing colouration can have potential costs in terms of reproductive interference since wing colouration is often involved in mate recognition. If the benefits of a shared antipredator defence outweigh the costs of reproductive interference, iridescence wing colouration will show convergence and alternative mate recognition strategies might evolve, such as chemical mate recognition. In allopatry, iridescence wing colouration is expected to diverge due to adaptation to different local conditions and no alternative mate recognition is expected.

Strengths:

(1) Using allopatric and sympatric (sub)species that are closely related is a powerful way to test evolutionary hypotheses.

(2) By clearly defining iridescence and measuring colour spectra from a variety of angles, applying different methods, a very comprehensive dataset of iridescence wing colouration is achieved.

(3) By experimentally manipulating wing coloration patterns, the authors show visual mate recognition for *M. h. bristowi* and could, in theory, separate different visual aspects of colouration (patterns VS iridescence strength).

(4) Measurements of chemical profiles to investigate alternative mate recognition strategies in case of convergence of visual signals.

Weaknesses:

In my opinion, studies should be judged on the methods and data included, and not on additional measurements that could have been taken or additional treatments/species that should be included, since in most ecological and evolutionary studies, more measurements or treatments/species can always be included. However, studies do need to ensure appropriate replication and appropriate measurements to test their hypothesis AND support their conclusions. The current study failed to ensure appropriate replication, and in various cases, the results do not support the conclusions.

First, when using allopatric and sympatric (sub)species pairs to test evolutionary hypotheses, replication is important. Ideally, multiple allopatric and sympatric (sub)species pairs are compared to avoid outlier (sub)species or pairs that lead to biased conclusions. Unfortunately, the current study compares 1 allopatric and 1 sympatric (sub)species pair, hence having poor (no) replication on the level of allopatric and sympatric (sub)species pairs.

Second, chemical profiles were only measured for sympatric species and not for allopatric (sub)species, which limits the interpretation of this data. The allopatric (sub)species could have been measured as non-coexistence "control". If coexistence and convergence in wing colouration drives the evolution of alternative mate recognition signals, such alternative signals should not evolve/diverge for allopatric (sub)species where wing colouration is still a reliable mate recognition cue. More importantly, no details are provided on the quantification of butterfly chemical profiles, which is essential to understand such data. It is unclear how the chemical profiles were quantified and what data (concentrations, ratios, proportions) were used to perform NDMS and generate Figure 5 and the associated statistical tests.

Third, throughout the discussion, the authors mention that their results support natural selection by predators on iridescent wing colouration, without measuring natural selection by predators or any other measure related to predation. It is unclear by what predators any of the butterfly species are predated on at this point.

To continue on the interpretation of the data related to selection on specific traits by specific selection agents: This study did not measure any form of selection or any selection agent. Hence, it is not known if iridescent wing colouration is actually under selection by predators and/or mates, if maybe other selection agents are involved or if these traits converge due to genetic correlations with other traits under selection. For example, Iridescent colouration in ground beetles has functions as antipredator defence but also thermo- and water regulation. None of these issues are recognized or discussed.

Finally, some of the results are weakly supported by statistics or questionable methodology.

Most notably, the perception of the iridescence coloration of allopatric subspecies by bird visual systems. Although for females, means and errors (not indicated what exactly, SD, SE or CI) are clearly above the 1 JND line, for males, means are only slightly above this line and errors or CIs clearly overlap with the 1 JND line. Since there is no additional statistical support, higher means but overlap of SD, SE or CI with the baseline provides weak statistical support for differences.

Regarding the assortative mating experiment, the results are clearly driven by *M. bristowi*. For *M. theodorus*, females mate equally often with conspecifics (6 times) as with *M. bristowi* (5 times). For males, the ratio is slightly better (6 vs 3), but with such low numbers, I doubt this is statistically testable. Overall low mating for *M. bristowi* could indicate suboptimal experimental conditions, and hence results should be interpreted with care.

Regarding the wing manipulation experiment, *M. theodorus* does not show a preference when dummies with non-modified wings are presented and prefers non-modified dummies over modified dummies. This is acknowledged by the authors but not further discussed. Certainly, some control treatment for wing modification could have been added.

Overall, the fact that certain measurements only provide evidence for 1 of the 2 (sub)species (assortative mating, wing manipulation) or one sex of one of the species (bird visual systems) means overall interpretation and overgeneralization of the results to both allopatric or sympatric species should be done with care, and such nuances should ideally be discussed.

The aim of the authors, "to investigate the antagonistic effects of selective pressures generated by mate recognition and shared predation" has not been achieved, and the conclusions regarding this aim are not supported by the results. Nevertheless, the iridescence colour measurements are solid, and some of the behavioural experiments and chemical profile measurements seem to yield interesting results. The study would benefit from less overinterpretation of the results in the framework of predation and more careful consideration of methodological difficulties, statistical insecurities, and nuances in the results.

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Author response:

Reviewer #1 (Public review):

(1) The broader significance of the findings needs to be better articulated. While the authors emphasize that comparing adaptive traits in sympatry and allopatry provides insights into selective processes shaping reproductive isolation and coexistence, it is unclear what key conceptual or theoretical questions are being addressed. Are these patterns expected under certain evolutionary scenarios? Have they been empirically demonstrated in other systems? The authors should explicitly state the overarching research question, incorporate some predictions, and better contextualize their findings within the existing literature. If the results challenge or support previous work, that should be highlighted to strengthen the study's importance in a broader context.

We thank the reviewer for their valuable feedback. We understand that the framing of the results and the discussion did not allow to highlight the broader significance of our findings. In the revised version of the manuscript, we will explicitly mention the theoretical questions asked and our hypotheses in the introduction, and better compare our results to pre-existing examples from the literature.

(2) The motivation for studying visual signals and mate choice in allopatric populations (i.e., at the intraspecific level) is not well articulated, leaving their role in the broader narrative unclear. In particular, the rationale behind experiments 1, 2, and 3 is not well defined, as the authors have not made a strong case for the need for these intraspecific comparisons in the introduction. This issue is further compounded by the authors' primary focus on signal evolution in sympatry throughout both the results and the

discussion. For instance, the divergence of iridescence in allopatry is a potentially interesting result. But the authors have not discussed its implications.

Overall, given that the primary conclusions are based on results and analyses in sympatry, the role of allopatric populations in shaping these conclusions needs to be better integrated and justified.

Without a stronger link between the comparative framework and the study's key takeaways, the use of allopatric populations feels somewhat peripheral rather than central to the study's aim.

Since the primary conclusions remain valid even without the allopatric comparisons, their inclusion requires a clearer rationale.

We recognize that the current manuscript places more emphasis on the sympatric Morpho population, and that the analysis and the discussion of the results regarding the allopatric Morpho population were underdeveloped. In the revised version, we plan to address this by (1) developing the rationale behind the male choice experiments performed on the allopatric population. We will argue that intraspecific comparison helps identify the traits involved in mate preference within species (iridescent color and/or wing pattern) and that those results can be compared to the interspecific mate choice results to identify the traits involved in species recognition. To explain the relevance of the comparison with the allopatric population, we will also (2) strengthen expectations on the effect of species interactions on the evolution of traits and mate recognition in sympatric populations vs. allopatric populations.

(3) While the authors demonstrate that iridescence is indistinguishable to predators in sympatry, they overstate the role of predation in driving convergence. The present study does not experimentally demonstrate that iridescence in this species has a confusion effect or contributes to evasive mimicry. Alternatively, convergence could result from other selective forces, such as signal efficacy due to environmental conditions, rather than being solely driven by predation.

We acknowledge that this study neither demonstrates that iridescence contributes to evasive mimicry nor that predation is the driver of the convergence in iridescence. We will tone down the interpretation of the results in the discussion and state that predation is not the only selective pressure that could have promoted a convergent evolution of iridescence in sympatric species, although this observation is consistent with the evasive mimicry hypothesis.

Reviewer #2 (Public review):

My only major comment concerns the authors' favoured explanation for aposematism (or evasive mimicry) for convergence among species, which is based upon the you-can't-catch-me hypothesis first presented by Young 1971. Although there is supporting work showing that iridescent-like stimuli are more difficult to precisely localize by a range of viewers, most of the evidence as applied to the Morpho system is circumstantial, and I'm not certain that there is widespread acceptance of this hypothesis. Given that the present study deals with closely-related (sub)species, one alternative explanation - a "null" hypothesis of sorts - is for a lack of divergence (from a common starting point) as opposed to evolutionary convergence per se. In other words, two subspecies are likely to retain ancestral character states unless there is selection that causes them to diverge. I feel that the manuscript would benefit from a discussion of this alternative, if not others. Signalling to predators could very well be involved in constraining the extent of convergence, but this seems a little premature to state as an up-front conclusion of this

work. There is also the result of a **dorsal** wing manipulation by Vieira-Silva et al. 2024 (<https://doi.org/10.1111/eth.13517>), which seems difficult to reconcile in light of this explanation. Whereas this paper is cited by the authors, a more nuanced discussion of their experimental results would seem appropriate here.

We thank the reviewer for their constructive comments on our manuscript. We appreciate the reviewer's concern regarding the way iridescence convergence between sympatric species is discussed in our manuscript, which aligns with similar concerns raised by Reviewer 1. We will improve the discussion on the different evolutionary forces that could have favored this convergent iridescent signal in sympatry to bring more nuance to the discussion.

Reviewer #3 (Public review):

First, when using allopatric and sympatric (sub)species pairs to test evolutionary hypotheses, replication is important. Ideally, multiple allopatric and sympatric (sub)species pairs are compared to avoid outlier (sub)species or pairs that lead to biased conclusions. Unfortunately, the current study compares 1 allopatric and 1 sympatric (sub)species pair, hence having poor (no) replication on the level of allopatric and sympatric (sub)species pairs.

We would like to thank the reviewer for their constructive feedbacks. We agree that replication is important to test evolutionary hypotheses and that our study lacks replication for allopatric and sympatric Morpho populations. Ideally, one would require several allopatric and sympatric replicates pointing respectively toward divergence and convergence of Morpho iridescence to conclude on the effect of species interaction in trait evolution. Our study is a first attempt at answering this question, covering few Morpho populations but proposing a broad assessment of iridescence and mate preference for those populations. We will make sure to mention this limitation more clearly in the revised version of our manuscript.

Second, chemical profiles were only measured for sympatric species and not for allopatric (sub)species, which limits the interpretation of this data. The allopatric (sub)species could have been measured as non-coexistence "control". If coexistence and convergence in wing colouration drives the evolution of alternative mate recognition signals, such alternative signals should not evolve/diverge for allopatric (sub)species where wing colouration is still a reliable mate recognition cue. More importantly, no details are provided on the quantification of butterfly chemical profiles, which is essential to understand such data. It is unclear how the chemical profiles were quantified and what data (concentrations, ratios, proportions) were used to perform NDMS and generate Figure 5 and the associated statistical tests.

We recognize that having the chemical profiles of the genitalia of the Morpho from the allopatric population would have made a stronger case arguing in favor of reinforcement acting on the divergence of the chemical compounds found on the genitalia of the sympatric Morpho species. Due to limited access to the biological material needed by the time of the chromatography, we could not test for lower divergence in the chemical profiles of allopatric Morpho butterflies. We will mention this limitation in the results, and clarify the protocol used to extract the chemical profiles, by mentioning the use of concentration data to generate Figure 5 and the associated statistical tests.

Third, throughout the discussion, the authors mention that their results support natural selection by predators on iridescent wing colouration, without measuring natural

selection by predators or any other measure related to predation. It is unclear by what predators any of the butterfly species are predated on at this point.

We will mention in the next version of the manuscript previous predation experiments performed on *Morpho* and other butterflies showing evidence that birds can be predators for those species. Those observations lead us to test for the putative effect of predation on the evolution of their color pattern, without directly testing predatory rates. We will make sure this information is transparent in the revised manuscript.

To continue on the interpretation of the data related to selection on specific traits by specific selection agents: This study did not measure any form of selection or any selection agent. Hence, it is not known if iridescent wing colouration is actually under selection by predators and/or mates, if maybe other selection agents are involved or if these traits converge due to genetic correlations with other traits under selection. For example, Iridescent colouration in ground beetles has functions as antipredator defence but also thermo- and water regulation. None of these issues are recognized or discussed.

We acknowledge that the lack of discussion on alternative evolutionary forces involved in the evolution of iridescence has been highlighted by all reviewers. We will discuss how environmental factors, genetic factors or the correlation with others traits as explanatory variables might explain the convergent signal of iridescence found in sympatric *Morpho* species, and not only focus on the putative effect of predation.

Finally, some of the results are weakly supported by statistics or questionable methodology. Most notably, the perception of the iridescence coloration of allopatric subspecies by bird visual systems. Although for females, means and errors (not indicated what exactly, SD, SE or CI) are clearly above the 1 JND line, for males, means are only slightly above this line and errors or CIs clearly overlap with the 1 JND line. Since there is no additional statistical support, higher means but overlap of SD, SE or CI with the baseline provides weak statistical support for differences.

We thank the reviewer for bringing interpretation issues concerning the chromatic distances of allopatric *Morpho* species measured with a bird vision model. We will make sure to bring nuance to the interpretation of this graph, and clearly mention in the figure's legend that the error bars represent the confidence intervals obtained after performing a bootstrap analysis.

*Regarding the assortative mating experiment, the results are clearly driven by *M. bristowi*. For *M. theodorus*, females mate equally often with conspecifics (6 times) as with *M. bristowi* (5 times). For males, the ratio is slightly better (6 vs 3), but with such low numbers, I doubt this is statistically testable. Overall low mating for *M. bristowi* could indicate suboptimal experimental conditions, and hence results should be interpreted with care.*

*Regarding the wing manipulation experiment, *M. theodorus* does not show a preference when dummies with non-modified wings are presented and prefers non-modified dummies over modified dummies. This is acknowledged by the authors but not further discussed. Certainly, some control treatment for wing modification could have been added.*

We recognize that the tetrad experiment results are mainly driven by *M. bristowi*'s behavior. This experiment would have benefited from more replicates. We will mention that the conclusions we draw for this experiment are mainly driven by male *M. bristowi* behavior, and that it is more difficult to test for assortative or disassortative mating in *M. theodorus*,

adding more nuance to our interpretation. We will also make sure to discuss further the effect of wing modification in the discussion.

Overall, the fact that certain measurements only provide evidence for 1 of the 2 (sub)species (assortative mating, wing manipulation) or one sex of one of the species (bird visual systems) means overall interpretation and overgeneralization of the results to both allopatric or sympatric species should be done with care, and such nuances should ideally be discussed.

The aim of the authors, "to investigate the antagonistic effects of selective pressures generated by mate recognition and shared predation" has not been achieved, and the conclusions regarding this aim are not supported by the results. Nevertheless, the iridescence colour measurements are solid, and some of the behavioural experiments and chemical profile measurements seem to yield interesting results. The study would benefit from less overinterpretation of the results in the framework of predation and more careful consideration of methodological difficulties, statistical insecurities, and nuances in the results.

Overall, we would like to thank all reviewers for their thorough assessment of our work. We understand that the imbalance between mate choice data, visual model data and chemical data only give us a partial assessment of species recognition in *Morpho* butterflies, thus requiring more precision in the interpretation and the discussion of our results. We will implement all the comments made by the reviewers in the next version of our manuscript.

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